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Is There Learnable Structure Beyond a Strong Additive Survival Model? A Locally Operated, Supervised Research Agent Tests It on Single-Visit Cohort Data

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Working draft, started 2026-10-07; last revised 2026-10-08 after the secondary estimands (sections 4.5, 4.6 and 5.5 to 5.10) were added to the first session of delegated experiments. Every numerical claim in this document is either reproduced by a command listed in the Reproducibility section or is explicitly marked as unmeasured. The related-work section covers the works cited and is not a survey.

Abstract

Deep survival models have beaten classical baselines where patients number in the hundreds of thousands and events in the billions. We ask what happens at the scale of one examination per person (about 56 thousand adults and 8.4 thousand deaths from US national survey cycles linked to mortality records), and we ask it with a method that makes the experiments themselves auditable: a locally served 27-billion-parameter coding agent carries out delegated tasks under the supervision of a stronger model, every run is kept in a hash-chained event log, and results with help are recorded apart from results without it. On the survival question, a regularised Cox model that is nonlinear in every continuous input (written by the agent) is the best model we measured by point estimate; it is not resolvable better than the same model without splines when all inputs are used (difference in integrated Brier score -0.00013 , 95% interval -0.00057 to $+0.00031$). A deep set encoder over the same inputs is worse than the spline Cox model by 0.0011 ($+0.0002$ to $+0.0020$) in a fold test; a pooled cluster bootstrap resolves that gap only at ten years, and the average of three training seeds closes it ($+0.0002$, -0.0008 to $+0.0012$). Learning curves over five cross-validation repeats show the Cox models flat from half of the training subjects and the more flexible learners still improving, so more data of this kind would help them and not the Cox models. The experiment that would test for structure beyond the additive model, a residual neural network, was not delivered by the agent in five attempts (two runs); written and run by the supervisor under a rule fixed beforehand, it did not improve on the spline Cox model (integrated Brier score -0.00008 , interval -0.00030 to $+0.00015$, against the 0.001 the rule required). The same single examination was then asked further questions under rules fixed beforehand: it predicts death by cause (the full input set beats the conventional risk factors for cardiovascular, cancer and other deaths), death with a diabetes mention on the certificate beyond who dies (and a hypertension mention not at all), expected time lived (calibrated within 0.05 years at ten years), eight prevalent conditions (AUROC 0.79 to 0.92 with the full input set, 0.67 to 0.85 without the inputs that define each), and undiagnosed disease only weakly (sensitivity 0.2 to 0.45 at 90% specificity; about nine of ten positive calls for diabetes are false); accelerometer summaries added a small, narrowly resolved gain to the ten-year Brier score, and a convolutional network on the minute series did no better than eight summaries. Which inputs the survival model relies on was then asked under a rule fixed beforehand: age dominates (a ratio of 26 between its 90th and 10th percentile), 38 of 67 inputs carry independent information, and the ranking survives the exclusion of early deaths (rank correlation 0.92); the directions are associations, and weight and ALT point the wrong way for advice because they mark frailty. On the agent question, two of four substantive tasks were accepted and two were not. Before the first could even be judged, four defects of the surrounding infrastructure (a reasoning block spending the output budget, greedy decoding repeating one action, a server context overflow, and an unreadable old log format) had to be found and repaired; two defects of the supervisor’s own acceptance checks surfaced later. Each repair has a regression specification. A candidate adapter trained on ten records from accepted runs was judged on eight held-out tasks under a rule fixed beforehand and rejected (0 of 8 against 1 of 8), leaving the model in use unchanged.

1. Introduction

Deep survival and event-sequence models have produced large gains over classical baselines where patients number in the hundreds of thousands to millions and timestamped events in the billions. Our own campaign works at a very different scale: one examination per person, about 8.4 thousand deaths, and a model of about 85 thousand parameters. Initial results showed that no baseline we tried was resolvable better than the neural model, and the neural model was not resolvable better than a regularised additive model.

We therefore reframe the research question from “which neural architecture is best on this data” to three separable questions:

1. **Nonlinear residual signal.** Does any learnable structure remain once a strong additive survival model has been fitted?
2. **Longitudinal signal.** Does repeated-measure information add predictive value beyond the current state?
3. **Sufficient data.** Is there enough data to learn what exists?

A single-visit cohort can answer only the first question, and this paper is organised around answering it with an experimental protocol that is fixed before the results are seen.

A second, methodological thread runs through the paper. The experiments are carried out by a locally served coding agent under the supervision of a stronger model, with every run recorded in an append-only trace. We report supervisor interventions separately from unaided results, because the claim of interest is how much of the work the local agent can do without help.

2. Related Work

Foundation models for time-to-event data. MOTOR pretrains a time-to-event transformer on tens of millions of patient records and billions of clinical events and reports gains in time-dependent concordance and label efficiency on downstream survival tasks (arXiv:2301.03150). SurvivEHR pretrains a competing-risks next-event model on 7.6 billion coded events from 23 million primary-care patients (Oxford and Birmingham, 2025, medRxiv). Delphi-2M trains a generative health-event transformer on 402,799 UK Biobank participants and, in a scaling experiment, finds that about two million parameters is optimal for that dataset (“Learning the natural history of human disease with generative transformers”, 2025). These results locate the regime where representation learning over event sequences pays: very many patients, many events per patient. Our cohort has one examination per person, so we test the weaker claim that nonlinear structure beyond a strong additive model exists at all.

Survival evaluation. We use the Aalen-Johansen estimator for observed cause-specific risk, inverse-probability-of-censoring weights for concordance and Brier scores (Uno et al.), and the corrected resampled t-test of Nadeau and Bengio for repeated cross-validation, whose variance inflation accounts for the overlap between training sets. Sample-size guidance for validating a survival prediction model depends on the distribution of predicted risks, the incidence and censoring, and the precision wanted (Riley et al., 2022, Statistics in Medicine); we do not claim a universal events threshold.

Neural networks and Cox regression. DeepSurv (Katzman et al., BMC Medical Research Methodology, 2018) replaces the linear predictor of the Cox model by a neural network and reports performance as good as or better than other survival models on its benchmarks. Kvamme, Borgan and Scheel (Journal of Machine Learning Research 20, 2019) propose Cox-based networks with a loss that scales to large data and find them highly competitive, typically the best in Brier score and binomial log-likelihood, on real data sets. Our residual network is of this family (a Cox predictor with a network added to the linear part), and it is asked a narrower question than either paper: whether it improves on a Cox model that is already nonlinear in every input, in a cohort of one visit per person.

Tabular deep learning. “On Embeddings for Numerical Features in Tabular Deep Learning” (Gorishniy, Rubachev and Babenko, NeurIPS 2022, arXiv:2203.05556) argues that embedding numerical features, by piecewise-linear encoding or periodic activations, is an underexplored freedom that lets simple MLP-like models perform on par with attention-based architectures and compete with gradient-boosted trees on some benchmarks. We did not use such embeddings: the inputs of the residual network are the standardised, spline-free columns of the shared preprocessing, and the spline Cox base already carries the per-input nonlinearity. An embedding-based network is a different experiment.

Biological age from NHANES. Levine et al. (“An epigenetic biomarker of aging for lifespan and healthspan”, Aging 10, 2018) built a phenotypic age from routine clinical biomarkers in NHANES III (9,926 adults with complete data and more than twenty years of mortality follow-up) and used it to derive an epigenetic predictor. Our mortality-equivalent age (section 5.7) is a different object: a re-expression of a survival model’s expected time lived against a reference life table, with its association to outcomes reported with a control that separates construction from information, and with no claim that it measures biological ageing.

Coding agents. SWE-bench (Jimenez et al., ICLR 2024, arXiv:2310.06770) evaluates language models on 2,294 real GitHub issues by running the tests of the repository after the model’s edit, and reports that its best model at the time resolved 1.96% of them. The loop of section 4.4 shares the principle of judging a patch by execution against checks the worker cannot edit; it is evaluated on four tasks of our own and nothing is claimed about its rate on any benchmark.

3. Data and Estimands

Cohort. The ten continuous cycles of the US National Health and Nutrition Examination Survey (1999 to 2018), linked to the public-use mortality file with follow-up to 31 December 2019. Every examined adult eligible for linkage is a subject: 56,253 subjects, 8,355 deaths. Each subject has one examination; the timeline holds what was measured or answered then, the eating-time features of the 24-hour dietary recalls, the ages at which conditions were first diagnosed and recalled body weights as history, and death in three competing cause groups (cardiovascular, cancer, other). A locked test set of 8,437 subjects was scored once under an earlier pre-registration and is not used here.

What the data cannot show. Death is the only outcome observed after the examination. Follow-up is administratively censored, so a cycle examined in 2015 is followed for about four years; a metric at horizon h is computed on the cycles whose survivors were all followed h years, and the 15-year metric rests on the 1999 to 2002 cycles

alone (1,583 subjects, 345 deaths in the primary analysis). The mortality file is perturbed by the data owner for some records. Ages are top-coded. Associations are estimated, not effects of changing a behaviour.

Estimands. For each subject the all-cause cumulative incidence by years 1 to 15, and the incidence of each cause where the model gives it. Discrimination is Uno’s concordance at 5 and 10 years; accuracy is the inverse-probability-of-censoring-weighted integrated Brier score over 0 to 15 years (IBS) and the Brier score at 10 years; calibration at 10 years is the weighted intercept and slope with an Aalen-Johansen observed risk per equal-weight risk group. Metrics are design-weighted with the survey weights; training is unweighted.

Folds. Five repeats of five-fold cross-validation over the non-locked subjects (25 folds), fixed in a frozen partition file. Every prediction used here is an out-of-fold prediction, produced by a model fitted on the other folds of its repeat.

4. Methods

One examination, many questions

NHANES 1999-2018 linked to mortality: 56,253 adults, 8,355 deaths, 5x5 cross-validation, rules fixed before results

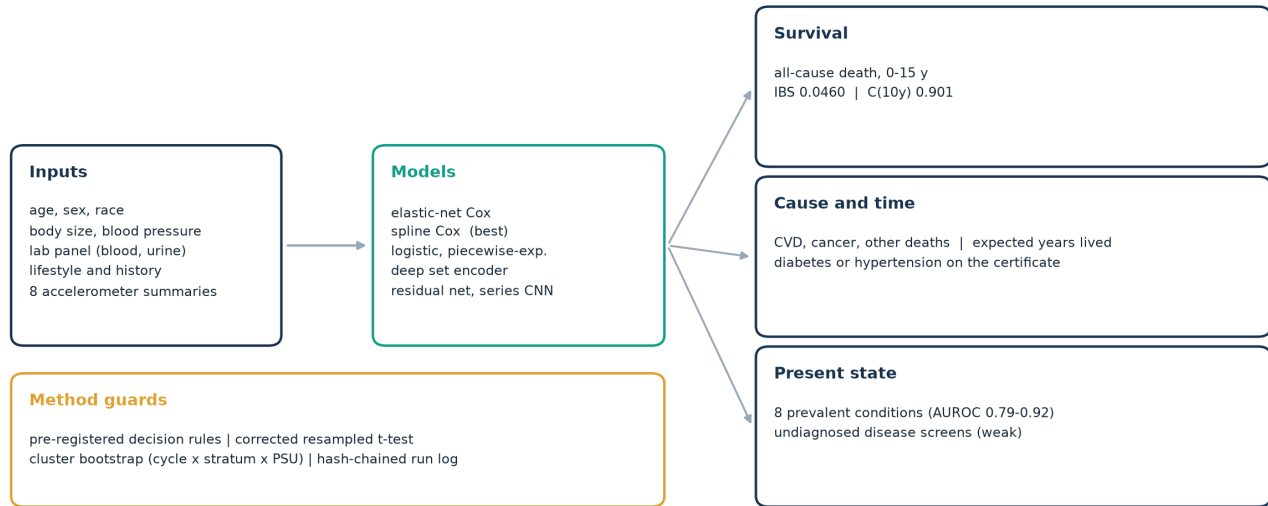


Figure 1: The study: inputs, models and the questions asked of one examination.

4.1 Models compared

All models predict the same cumulative-incidence curves from the same inputs and are evaluated by the same code.

- **age-sex** and **standard**: additive piecewise-exponential hazard models on age and sex, or on the conventional risk factors (reference points).
- **additive**: an additive proportional-hazards model (a piecewise-exponential generalised additive model) on every input; no interactions. About 16 thousand parameters.
- **horizon**: a set encoder over the subject’s tokens (variable identity, value, time) with attention, trained with the same likelihood and a masked value objective; about 86 thousand parameters at its default.
- **Conventional baselines** fitted in Python on the same folds: elastic-net Cox on the standard or all inputs, cause-specific Cox, inverse-probability-weighted logistic models, gradient-boosted survival trees.
- **Spline Cox** (added during this work): the elastic-net Cox model with a cubic B-spline basis on every continuous input, to establish the nonlinear additive ceiling.
- **Ensembles and blends**: equal-weight averages of members’ predicted curves.

4.2 Evaluation protocol

Two ways of estimating the difference between two models were planned. The stronger is on the pooled out-of-fold predictions of one repeat (each subject appears once) with a paired bootstrap: subjects, or PSUs within strata, are resampled with replacement and both models are scored on the same resample. The other is the corrected resampled t-test over the 25 folds (which inflates the variance for the overlap between training sets), never an uncorrected test. For the primary comparisons **only the corrected t-test was run**: the bootstrap was specified as a task for the agent and was not run for lack of time (section 7). The secondary estimands of sections 5.6, 5.7 and 5.9 use a cluster bootstrap (cycle, stratum, PSU) on pooled out-of-fold predictions of one repeat. Model choices that used the folds (a capacity sweep, say) are made on repeat 0 and confirmed on repeats not used to choose.

4.3 Pre-registered decision rule

A deep model is called useful on this cohort only if it improves IBS by at least 0.001 reproducibly across seeds and folds, with a paired confidence interval that excludes zero. Failure to meet the rule is reported as a negative result and is not read as proof of equivalence.

4.4 The research agent

The experiments are carried out by a coding agent, the *loop*, that runs on a locally served language model and is supervised by a stronger model. The design goal is that a supervisor can audit and reproduce what the agent did, and that results with help are never mixed with results without it.

Task contract. A run starts from a contract that is written before the first model call and never changed: the task text, the repository and its baseline revision, the acceptance checks, a list of protected paths, the limits, the model reference, whether models reached over an API were allowed, and the digest of the system prompt in effect. Acceptance checks are shell commands chosen by the supervisor and kept outside the checkout the worker can edit; protected paths (for example the repository’s own tests) may not change, and a candidate that touches one is rejected even if every check passes.

Isolation and tools. The worker edits a detached git worktree of the baseline revision, never the repository named by the task. It is given the file, search and shell tools of the agent runtime it is built on, confined to that checkout by the runtime’s path jail; tools that reach the network, delegate to another model, or switch the model are withheld. The checkout is isolation from accidents and not a sandbox: the shell runs as the user.

Limits. One attempt is bounded by wall-clock time, generated tokens and a cap on tool calls (the runtime has no run option for the last, so the loop counts tool requests and cancels the run at the cap); the run is bounded by total time, the number of attempts, bounded retries of a model call that failed in transit, and, for a remote model, a cost cap. A remote call whose price the provider does not report is counted and never treated as free.

Judging and retry. After an attempt the loop reads back from git what changed (a patch, and for each file its status and content hashes before and after), checks the protected paths, and runs the acceptance checks in the candidate checkout. A failed attempt is classified (acceptance failed, protected path touched, tool-call cap, time, token, provider, cancelled), and the next attempt starts from a clean checkout and is shown the failure and the tail of the failing output, never a patch.

Trace. Every run keeps an append-only event stream, written in the agent runtime’s own hash-chained log format (each line carries the hash of the one before, so an edited, reordered, inserted or removed event is reported when the stream is read, and writers in several processes extend one chain). An event has a schema version, run, attempt, an id that only grows, timestamp, parent, type and data, with credentials redacted before anything is written and payloads over a size bound stored once by content address. The chain detects accidental damage and edits that do not recompute every later hash; it is not a defence against someone who can rewrite the whole file. The runtime’s own trajectory of each attempt is persisted beside it. A gap in what could be observed is an event of its own. The run records which subagent, skill, command and project-context definitions were in effect, found by the runtime’s own discovery rules, as a digest in the contract and a **definitions** event, and flags a change on resume. Checkpoints are written after every attempt; a resumed run does not trust the checkout an interrupted attempt left, resets it to the baseline and records that it did.

Stopping a worker that is going nowhere. An attempt is stopped as *stagnation* when one call (same tool, same arguments) has returned the same answer a set number of times (default four); a repeated call whose answer changes, such as a test run after an edit, is not counted. The retry is told which call looped, never a patch. This

guard exists because of what we observed (section 5.1): a 27-billion-parameter model decoded greedily repeated one probe command about forty times and wrote nothing.

Accounting of help. A run given any supervisor hint is recorded as assisted. Unaided means no hint, no remote model and no supervisor repair during the attempt. Remote models require an explicit opt-in flag; a model served by the model engine on this machine counts as local. Settings the supervisor chooses (sampling temperature, output and tool-call budgets) are configuration, written in the run, and do not make a run assisted. In this paper the two accepted tasks were unaided in that sense (no hint); the continuation of the residual-network task was assisted (it was given a hint about a library call) and is labelled so. The infrastructure repairs of section 5.1 were made by the supervisor between runs, not during an attempt.

4.5 Secondary estimands and their pre-registered rules

Added 2026-10-08, before any number for these estimands was computed. They use the same cohort, folds and corrected resampled t-test as the primary question. All five are secondary: none changes the rule of section 4.3.

Id	Estimand	Rule fixed in advance
T1	Cumulative incidence of death by cardiovascular, cancer and other causes at 5, 10 and 15 years	A model beats the conventional risk-factor model for a cause iff the 10-year cause-specific Brier difference has a corrected 95% interval below zero. Calibrated iff the bootstrap observed-over-expected interval contains 1 and the slope lies in [0.8, 1.25]
T2	Cumulative incidence of death with diabetes or hypertension listed anywhere on the certificate (the linked file’s multiple-cause flags), with death without the flag competing	Models are ranked only with at least 100 flagged events at the horizon; below that, descriptive only. A flag model adds something only if it beats the all-cause risk used as a ranker, on time-dependent AUC, with an interval excluding zero
T3	Restricted mean survival time to 10 years (15 on the 1999-2002 cycles), and the age at which a reference life table gives the same time (mortality-equivalent age)	Calibrated iff the mean absolute gap over risk groups is at most 0.25 years at 10 years and the slope lies in [0.9, 1.1]
T4	Prevalent conditions at the examination: diabetes, hypertension, kidney markers, anaemia, high cholesterol, osteoporosis, depression, sleep problem	Useful iff AUROC exceeds the age-sex model by at least 0.02 with an interval excluding zero. Two input sets are reported side by side: all inputs except those that define the label, and non-definitional inputs only (demographics, body measures, recalled weights, smoking, alcohol, eating times)
T5	Undiagnosed disease among those not reporting the diagnosis	A screening value is claimed only if sensitivity at 90% specificity beats a model of age and body-mass index with a bootstrap interval excluding zero. Inputs are the non-definitional set only

What each estimand does not claim. A certificate mention is not a diagnosis and is under-reported; records without multiple-cause data are excluded and counted; there are few events, so intervals are wide and no subgroup claim

is made. Mortality-equivalent age is a re-expression of one model against one reference table, not a measure of biological ageing, and its association with death is partly by construction; only the cause-, flag- and condition-specific associations are informative. A prevalent condition at the same examination is a classification, not a forecast, and the labels derived from a measurement are definitional for the inputs that contain it. “Undiagnosed” is one measurement above a threshold plus recall of no diagnosis, not a confirmed diagnosis. Eight conditions and three causes are examined; all are reported and none is claimed beyond its rule. Label definitions are in Appendix C.

4.6 Accelerometer summaries as an input block (pre-registered 2026-10-08)

Written before the accelerometer files were opened. The 2003-2004 and 2005-2006 cycles carried a hip-worn uniaxial accelerometer for seven days, recorded per minute. Daily summaries are derived by a fixed recipe: non-wear is a run of at least 60 minutes of zero counts allowing at most two minutes of counts up to 100 inside it; a day is valid with at least 600 minutes of wear; a subject is included with at least four valid days. Per subject, over valid days: mean counts per wear minute, minutes per day at or above 2,020 counts (moderate to vigorous activity), minutes per day below 100 counts (sedentary), minutes per day in between (light), the highest 30-minute mean count, the probability that a sedentary minute is followed by an active one (a fragmentation measure), the number of valid days and the wear time.

The question is whether these summaries add to the same model without them. It is asked on the subjects of the two cycles with a valid recording, with the folds of the partition restricted to them: the spline Cox model on all inputs, against the same model with the summaries appended, trained and scored on the restricted folds. The block is called useful only if the 10-year Brier score falls, with the corrected resampled t-test interval of the paired difference below zero; Uno’s concordance at 10 years is reported beside it. The restricted subsample is small (a few thousand adults and a few hundred deaths by 10 years), so a failure to meet the rule is a statement about this sample, not about the signal. A model that reads the minute sequences directly (rather than summaries) is not compared.

4.7 The residual network and its ablations (pre-registered 2026-10-08)

Written before the model was run. The question of section 1 (does structure remain beyond a strong additive model?) is asked with a residual network: the linear predictor is the fitted spline Cox predictor plus the output of a small multilayer perceptron on the preprocessed inputs, whose last layer starts at zero, so training begins exactly at the spline model. The base coefficients are fitted on the training subjects as in section 4.1; the network is trained on the Cox partial likelihood with the base predictor as an offset, by Adam with weight decay and dropout. The number of epochs is chosen on a validation share held out of the training subjects, among epochs that include epoch zero (the spline model itself), and the model is then refitted on all training subjects for that many epochs. Every hyperparameter other than width and depth is fixed in advance (learning rate 0.001, weight decay 0.0001, dropout 0.1, at most 300 epochs, full batch).

The capacity sweep is the fixed list of widths 16, 64 and 256 with one hidden layer, and 64 with two and three. It is scored on repeat 0 (5 folds) by the integrated Brier score against the spline Cox model; the best configuration by that score is the headline configuration. It is then confirmed on repeats 1 to 4 (20 folds) alone. The decision rule is that of section 4.3: the residual network is useful only if the integrated Brier score improves by at least 0.001 with a corrected resampled t-test interval of the paired difference excluding zero, on the confirmation folds. A failure is reported as a negative result for this network, data size and training recipe, not as proof that no structure exists.

Feature-block ablations ask where any gain comes from: the same network (the headline configuration) is given only one block of inputs at a time, the base model unchanged. The blocks are the examination measurements (body, blood pressure and laboratory values), the eating-time features, the recalled diagnosis ages and weights, and the questionnaire answers. A block is reported by its paired difference to the spline model on the confirmation folds; no rule is attached to the ablations, which are descriptive. A shuffled-outcome control (the network trained on outcomes permuted among the training subjects) must not improve on the spline model; if it does, the training procedure leaks.

4.8 A model of the accelerometer series against the summaries (pre-registered 2026-10-08)

Written before any sequence model was run. Section 4.6 asked whether eight daily summaries add to the other inputs. This asks whether the minute series carries more than the summaries do. Each subject’s recording is reduced

to a fixed grid: seven days of ten-minute bins (1,008), with two channels, the logarithm of one plus the mean counts per worn minute in the bin and the share of its minutes worn. A residual network of the same kind as section 4.7 is built on it: the linear predictor is a spline Cox predictor plus the output of a small convolutional network on the grid (two strided convolutions, global mean and maximum pooling, a linear head that starts at zero), trained by the same procedure, with the number of epochs chosen on a validation share. Four arms are scored on the restricted folds of section 4.6: the base alone; the base with the summaries; the base with the series network; the base with the summaries and the series network. The series network is useful beyond the base only if the 10-year Brier score falls with a corrected interval below zero against the base, and is better than the summaries only if it does so against the base with the summaries. The sample is the same few thousand adults, so a failure is a statement about this size of sample.

4.9 Which inputs the model relies on (pre-registered 2026-10-08)

Written, and the code committed (`baselines/drivers.py`, commit 1053873), before any importance was computed. The question is which of the examination’s inputs the survival model uses and in which direction, on repeat 0 of the folds. The model is the spline Cox model on every input (section 4.1). The importance of an input is the fall in the held-out Breslow partial log-likelihood per event, in nats, when that input’s values are permuted among the fold’s test subjects (mean of ten permutations). An input carries independent information if the mean fall over the five folds is at least 0.001 and the fall is positive in every fold. The direction is the hazard ratio between the 90th and the 10th percentile of a numeric input, or between each level and the most common level of a categorical one (levels with at least 2% of the test subjects), as the geometric mean over test subjects of the model’s own ratio with only that input set. A numeric input missing for more than 20% of the subjects gets no ratio, because setting it for everyone would invent a condition. As a sensitivity analysis against illness that is already present at the examination, the same analysis is repeated without the subjects who die in the first two years; the two importance rankings are compared by rank correlation and by the overlap of their ten most important inputs. Importance and direction describe the fitted model’s use of an input. They are associations and do not estimate what changing the input would do.

5. Results

Numbers are means over the 25 cross-validation folds unless a repeat is named; differences are paired on identical folds and carry the corrected resampled t-test interval. Lower is better for the integrated Brier score (IBS, 0 to 15 years) and the Brier score at 10 years; higher is better for Uno’s concordance (C) at 10 years.

5.1 Bringing the agent up: what had to be repaired before a model could be judged

The first delegated task (add a Cox model that is nonlinear in each continuous input) was attempted on a locally served 27-billion-parameter model (Q8_0 weights, served by the model engine on this machine, no remote call). The first attempts did not measure the model; they exposed defects of the infrastructure, each repaired, covered by a specification and rerun:

Finding	What was observed	Repair
F-012	Two attempts ended at the output-token cap with no file written: the model reasoned before every reply (5 to 13 thousand output tokens a call, empty visible text)	a model served on this machine is asked for no reasoning block unless <code>--thinking</code> is given
F-014, F-015	With the reasoning block off, the model repeated one probe command about forty times and wrote nothing; the attempt ended at the tool-call cap. At a sampling temperature of 0.7 the same task edited files within five minutes: greedy decoding was the cause	a stagnation stop (one call returning the same answer four times) with feedback naming the call; <code>--temperature</code> and <code>--thinking</code> are recorded flags

Finding	What was observed	Repair
F-016	A follow-up round at a conversation of 58 thousand tokens was refused by the server (prompt plus requested reply above its context)	the provider is built with the context window the server reports, so replies are bounded by the room left
F-013	The new hash-chained event stream could not read runs recorded before it	legacy streams are read and migrated on resume
F-017	My own acceptance script chained two README checks with <code>&&</code> under <code>set -e</code> , so the requirement was not checked and an accepted patch lacked it	the check was corrected; the README item was carried to a later task

With these in place the first task was accepted on its second attempt, with no hint and no remote model. The first attempt wrote a model that refit its spline knots on whatever rows it was handed, including the held-out rows; the supervisor’s hidden check (predictions for one held-out row must not change when the other held-out rows are replaced by extreme values) failed with the difference in the predictions, and the second attempt fitted the knots on the training rows only. A follow-up task, accepted on its first attempt, added two specification tests, one of which fails if the spline model is replaced by the linear one (checked by mutation). A third task, a learning-curve fitting tool, was not delivered: two attempts ended at the tool-call limit and the time limit, and the third was cancelled by the supervisor for budget. This is recorded as a failure of the agent on a task of that size, not as a result about learning curves; the learning-curve analysis reported below was made by the supervisor and has since been written into the repository.

Auditing the research agent

A local 27B coding agent worked under a supervising model; every run sits in a hash-chained log

Substantive tasks



2 accepted, 2 not accepted

Held-out adapter test (rule fixed beforehand)



Adapter rejected; model in use unchanged.

Infrastructure defects repaired first

- reasoning block spending the output budget
- greedy decoding repeating one action
- server context overflow
- unreadable old log format

Two defects in the supervisor’s own acceptance checks surfaced later. Each repair has a regression specification.

Figure 2: The agent’s tasks and the defects found in its surroundings (section 5.1).

5.2 The nonlinear additive ceiling

model	IBS 0-15 y	Brier 10 y	C 10 y	calibration slope 10 y
age and sex only	0.05311	0.06437	0.8554	1.04

model	IBS 0-15 y	Brier 10 y	C 10 y	calibration slope 10 y
conventional risk factors (additive)	0.04952	0.05946	0.8803	1.09
elastic-net Cox, all inputs	0.04613	0.05350	0.8986	1.08
spline Cox, all inputs	0.04600	0.05346	0.9007	1.08
(agent-authored) inverse-weighted logistic, all inputs	0.04605	0.05351	0.8999	1.02
additive piecewise-exponential model, all inputs	0.04649	0.05448	0.9000	1.09
deep set encoder (horizon), all inputs	0.04711	0.05488	0.8969	0.98

Allowing the Cox model to bend in each continuous input did not resolvably change the integrated Brier score when all inputs are used (spline minus linear, -0.00013, corrected 95% interval [-0.00057, +0.00031]), although it raised concordance at 10 years by +0.0021 [+0.0007, +0.0035]. With only the conventional risk factors the same change helps: -0.00062 [-0.00106, -0.00019]. The nonlinear additive model is the strongest model in the table by point estimate, and the deep set encoder is now resolvably worse than it: +0.00111 [+0.00019, +0.00204] in IBS ($p = 0.02$) and +0.00142 [+0.00057, +0.00228] in the Brier score at 10 years. Under the pre-registered rule (section 4.3) the deep encoder is therefore not useful on this cohort: it does not improve on the nonlinear additive baseline, it is slightly worse.

The protocol preferred a pooled analysis with a cluster bootstrap to the fold t-test (section 4.2); it was run afterwards (**lifecourse bootstrap**) on the pooled out-of-fold predictions of repeat 0, each subject once, resampling cycle-stratum-PSU clusters 1,000 times. It agrees where the two have the same data: spline minus linear Cox, -0.00011 [-0.00049, +0.00024] in the integrated Brier score (8,976 subjects whose cycles reach 15 years) and -0.00001 [-0.00035, +0.00036] at ten years (23,097). Against the spline Cox, the additive piecewise-exponential model is +0.00054 [+0.00001, +0.00107] and +0.00108 [+0.00055, +0.00161] (integrated, ten years), and the deep set encoder +0.00071 [-0.00011, +0.00155] and +0.00106 [+0.00031, +0.00184]. The deep encoder’s integrated gap of 0.0011 in the fold analysis is therefore smaller and unresolved in this one; at ten years it is resolved in both.

Seeds and an ensemble. The deep encoder of the table is the **horizon** arm trained once. Training the default recipe again with three seeds on all 25 folds (the recipe path of section 5.3, one seed each) gives integrated Brier differences from the spline Cox model of +0.00096 [-0.00013, +0.00205], +0.00093 [-0.00044, +0.00230] and +0.00136 [+0.00021, +0.00250]: a gap of about 0.001 that one seed in three resolves, with a seed-to-seed spread of 0.0004. The equal-weight average of the three seeds’ predicted curves is +0.00018 [-0.00081, +0.00118]: indistinguishable from the spline Cox model, and not ahead of it, which the rule would require by 0.001. At ten years the ensemble is +0.00047 [-0.00026, +0.00120] by the fold test and +0.0003 to +0.0007 by the pooled bootstrap on each repeat (resolved on one repeat of five). So the deep encoder’s deficit is mostly variance between training runs, which averaging removes, and no average of seeds beats the additive model.

5.3 Learning curves

Mean integrated Brier score over all 25 folds (five repeats of five folds) when a share of each fold’s training subjects is used, the test folds identical. The subsamples are nested (a subject kept at a smaller share is kept at every larger one) and one training seed is used for the deep recipes. The analysis is **baselines/curves.py**; an earlier version of this table, on repeat 0 only, came from a script outside the repository.

share of training subjects	elastic-net Cox	spline Cox	inverse-weighted logistic	additive PE model	deep set encoder
10%	0.04785	0.04818	0.04891	0.05241	0.05325
25%	0.04702	0.04716	0.04755	0.04948	0.04986

share of training subjects	elastic-net Cox	spline Cox	inverse-weighted logistic	additive PE model	deep set encoder
50%	0.04633	0.04649	0.04636	0.04748	0.04780
75%	0.04618	0.04604	0.04620	0.04685	0.04727
100%	0.04613	0.04600	0.04605	0.04645	0.04696

The Cox and logistic models are nearly flat from half the data on: doubling the sample from 50% to 100% lowers the error by 0.0002 (elastic-net Cox), 0.0005 (spline Cox) and 0.0003 (logistic), and going from 75% to 100% by 0.00005, 0.00004 and 0.00015. The piecewise-exponential additive model and the deep encoder are still falling (0.0010 and 0.0008 from 50% to 100%; 0.0004 and 0.0003 from 75% to 100%), so the gap between the learners narrows with data: the encoder is 0.0013 behind the spline Cox at half the data and 0.0010 at all of it.

Power-law fits $e_{\text{inf}} + a N^{-b}$ to the means, with percentile intervals from resampling the 25 folds:

model	asymptote e_{inf} [95% CI]	exponent b [95% CI]
elastic-net Cox	0.0450 [0.0412, 0.0462]	0.42 [0.14, 0.75]
spline Cox	0.0431 [0.0283, 0.0453]	0.25 [0.05, 0.52]
inverse-weighted logistic	0.0442 [0.0409, 0.0457]	0.43 [0.21, 0.67]
additive PE model	0.0423 [0.0380, 0.0443]	0.40 [0.24, 0.55]
deep set encoder	0.0444 [0.0420, 0.0461]	0.56 [0.40, 0.76]

With five repeats the exponents of the flexible learners are determined (the encoder’s 0.56 is the steepest) and the asymptotes of the two Cox-family fits still are not: their intervals reach from far below to about the current error. The fitted asymptotes of the additive model and of the encoder lie below the Cox models’ full-data error of 0.0460 (the additive model’s whole interval does; the encoder’s reaches it), which is what a larger sample of this kind would test. An extrapolation from five points is a description of a curve and a hypothesis for the next data set, not an estimate of what more subjects would give.

5.4 Learning from accepted runs, and a candidate that is rejected

The loop keeps a training record only from a run that was accepted, unaided and on a local model, and only from the accepted attempt (a rejected patch is not a target). Of 22 runs offered, 9 produced a whole-conversation record and 10 a patch-form record (the rest were rejected, errored or cancelled; the manifest names the reason for each). Seven of them are the same task, so the set is small and not diverse.

The first training attempt did not run: the trainer refused the whole- conversation records because the chat template of the student (Qwen3) renders an assistant turn differently when it is the last message, so no loss mask can be fixed for a multi-step tool conversation (F-021; the trainer states this limitation and fails loudly). Trimming stray whitespace from tool-only turns (F-020) was necessary and not sufficient. A primitive for exact boundaries in such templates is missing in the model engine and is recorded as open there. The loop therefore writes, on request, one exchange per accepted run: the task as it was given and the accepted patch as a single assistant turn.

On those 10 records (8 trained on, 2 held out) the student, an 8-billion- parameter model run in-process on the GPU, was fine-tuned with a rank-16 low-rank adapter for 30 optimiser steps. The training loss fell from 1.41 to 0.04, a sign of memorising eight records; the held-out loss fell from 1.216 to 1.193. The adapter was written (a file of 175 MB, with its digest) and registered as a *candidate*, not as the model in use. It was then loaded by eight fresh loop processes (the run’s `model_selected` event names the adapter) and run, with the model in use, over the same eight seeded tasks of families absent from the training data, one attempt each, under identical limits. The rule had been fixed in code before any result: at least eight paired tasks, the candidate must solve more of them, and a one-sided sign test over the tasks only one of the two solved must give p at most 0.10. The model in use solved 1 of 8; the candidate solved 0 of 8 (one task lost, none gained, $p = 1.0$). The candidate was rejected with that record, the model in use stayed `local:Qwen/Qwen3-8B`, and asking to roll back reported that there was no adapter in use to roll back from.

This proves the mechanics: a real forward and backward pass, an optimiser step, a saved and reloaded adapter, an evaluation on held-out tasks under a declared rule, a rejection that leaves the previous version in use. It does

Using all examination inputs beats the conventional score

Integrated Brier score over 0-15 years, lower is better; 25 held-out folds

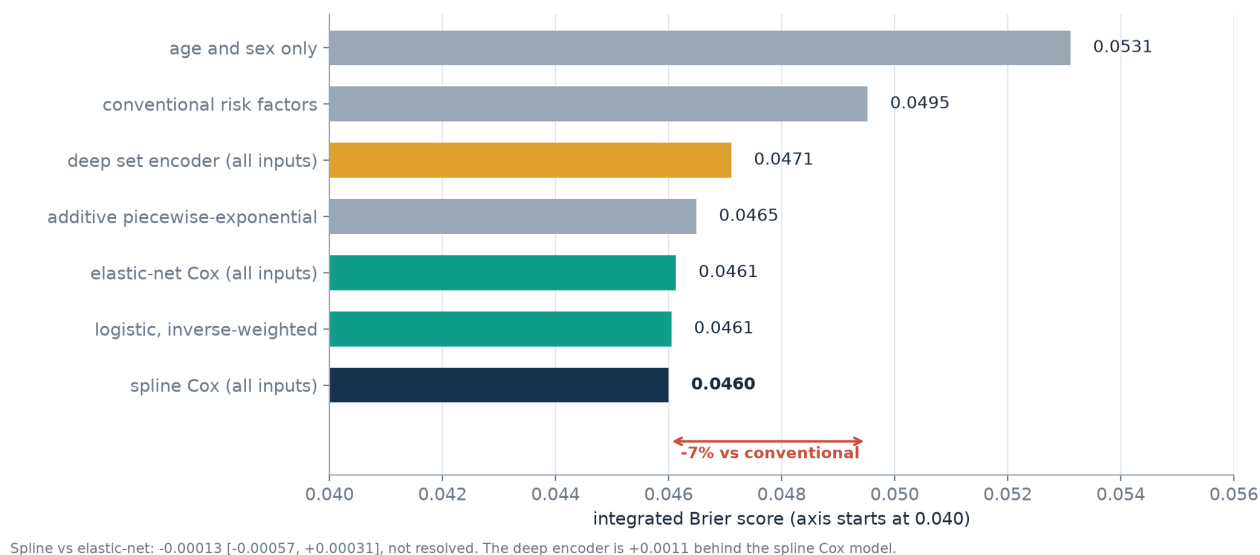


Figure 3: Integrated Brier score of each model (section 5.2).

The classical models are flat; the flexible ones still improve

Mean integrated Brier score vs share of training subjects, 25 folds, identical test folds

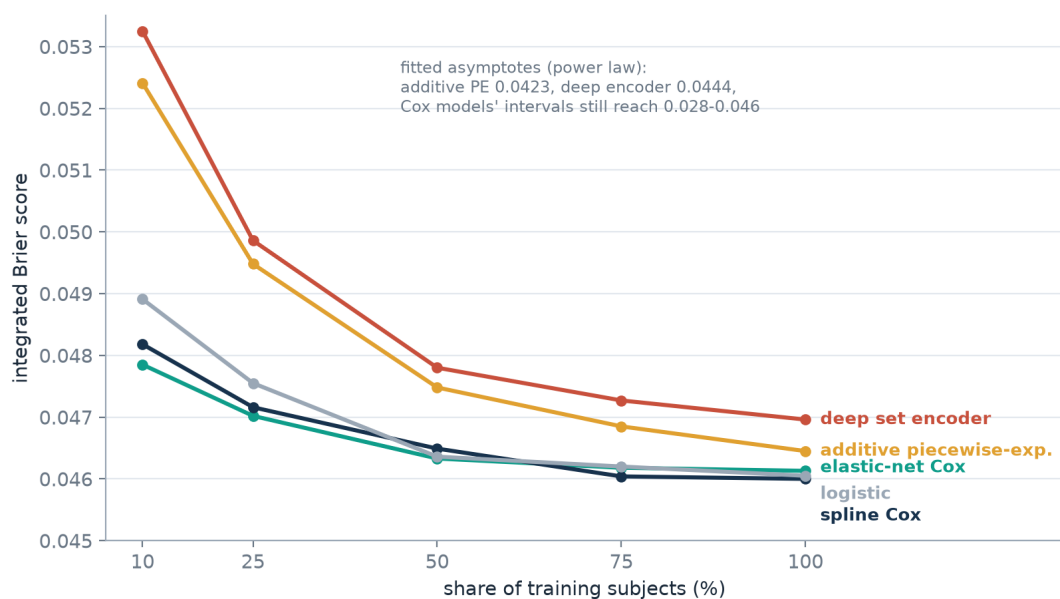


Figure 4: Learning curves (section 5.3).

not show that the loop learns: the student’s failures at baseline were malformed `edit_file` diffs repeated until the stagnation stop, and an adapter trained to write patches in prose is not a remedy for that. The baseline is one success in eight, so a gain would have needed to be large to be visible.

5.5 Secondary estimand T1: death by cause

Rules fixed in section 4.5 before the numbers existed. The models are three cause-specific Cox models fitted per cause on the same inputs as the all-cause baselines: age and sex alone, the conventional risk factors (**standard**), and every input (**all**). Each is scored on the 25 folds, by cause, at 5, 10 and 15 years, with the other causes competing. A horizon is scored only on the cycles whose follow-up reaches it, so the 15-year rows rest on the earliest cycles. The table gives the 10-year row; all rows, with the 5- and 15-year horizons, are in the run output (`m6/t1-causes.md`).

Cause (events per fold)	Model	Brier	Index of prediction accuracy	AUC	O/E	Slope	Brier difference to standard
Cardiovascularage, sex (213)		0.02685	0.124	0.906	0.963	1.06	+0.00090 [+0.00043, +0.00137]
	standard	0.02594	0.153	0.926	0.995	1.09	reference
	all	0.02522	0.177	0.938	0.984	1.08	-0.00072 [-0.00113, -0.00032]
Cancer (147)	age, sex	0.02268	0.043	0.862	0.948	1.10	+0.00030 [+0.00021, +0.00039]
	standard	0.02238	0.056	0.880	0.946	1.15	reference
	all	0.02190	0.076	0.894	0.961	1.10	-0.00048 [-0.00069, -0.00027]
Other (293)	age, sex	0.03886	0.120	0.857	0.922	1.04	+0.00087 [+0.00049, +0.00124]
	standard	0.03799	0.140	0.879	0.954	1.05	reference
	all	0.03618	0.181	0.911	0.942	1.07	-0.00181 [-0.00249, -0.00113]

Brackets are the corrected resampled t-test interval over the 25 folds. AUC is the time-dependent AUC at 10 years with event-free subjects as controls; O/E is observed over expected; the index of prediction accuracy is one minus the Brier score over that of a constant prediction.

By the rule of section 4.5, **all** beats **standard** for each of the three causes at 10 years: the interval of the Brier difference lies below zero in every case. The gain is largest for the heterogeneous residual group (-0.0018) and about half as large for cardiovascular (-0.0007) and cancer (-0.0005) death. At 15 years the cardiovascular interval touches zero ([-0.00135, +0.00007]) and the cancer AUC interval spans it, with 82 to 124 events per fold, so the rule is met for the other combinations and not for those two.

The calibration clause of the rule, tested afterwards on the pooled out-of-fold predictions of repeat 0 (23,097 subjects, 1,000 cluster resamples, the censoring distribution re-estimated on each), asks that the interval of observed over expected contain 1 and the slope lie in [0.8, 1.25]:

Cause (events)	Model	O/E [95% CI]	Slope [95% CI]	ICI	E90	Calibrated
Cardiovascular (1,067)	age, sex	0.962 [0.897, 1.033]	1.054 [0.981, 1.135]	0.0037	0.0127	yes
	standard	0.996 [0.933, 1.065]	1.081 [1.012, 1.152]	0.0027	0.0072	yes
	all	0.980 [0.918, 1.047]	1.083 [1.018, 1.151]	0.0031	0.0049	yes
Cancer (735)	age, sex	0.950 [0.847, 1.049]	1.087 [1.002, 1.191]	0.0021	0.0032	yes
	standard	0.947 [0.849, 1.046]	1.129 [1.044, 1.227]	0.0026	0.0053	yes
	all	0.960 [0.862, 1.053]	1.082 [1.005, 1.168]	0.0025	0.0041	yes
Other (1,464)	age, sex	0.924 [0.852, 0.988]	1.014 [0.939, 1.096]	0.0087	0.0214	no
	standard	0.958 [0.890, 1.020]	1.022 [0.949, 1.100]	0.0044	0.0105	yes
	all	0.943 [0.883, 1.002]	1.053 [0.991, 1.116]	0.0048	0.0090	yes
All causes (3,266)	age, sex	0.942 [0.904, 0.982]	1.042 [1.000, 1.088]	0.0120	0.0326	no
	standard	0.967 [0.932, 1.002]	1.078 [1.035, 1.124]	0.0068	0.0144	yes
	all	0.958 [0.928, 0.990]	1.094 [1.055, 1.136]	0.0094	0.0132	no

The full-input model is calibrated for each of the three causes. Summed over causes it predicts about 4% more deaths by ten years than occur (O/E 0.958, interval just below 1), and so does the age-sex model (0.942): the distinction is that the intervals of the cause rows, with fewer events, are wider and contain 1. ICI, E50 and E90 are the mean, median and 90th percentile of the absolute gap between the predicted risk and a smoothed observed risk (a local-linear smooth of the censoring-weighted outcome on the weighted rank of the prediction, a window of 30% of the weight): the typical gap is under 0.5 percentage points for every cause and 0.9 for all causes, and no model's 90th-percentile gap exceeds 2.2 points except age and sex for all causes (3.3). Slopes above one say the predictions are not spread out enough, by 5% to 13%. No criterion here uses the two-parameter recalibration intercept. It is read at a predicted risk of one half, so for cohorts whose risks are a few per cent it mostly restates the slope error (a prediction squeezed towards its mean has an intercept of 1.5 and a calibration-in-the-large of 0.15 in the specification that shows this); the release gate of the system the study ran on was changed to judge calibration in the large for that reason.

Reading the numbers. The high AUCs are largely age: age and sex alone reach 0.86 to 0.91, and the other inputs add 0.03 to 0.05. The index of prediction accuracy shows how much is left in absolute terms: cancer death is the least predictable cause (index 0.076 with all inputs against 0.177 for cardiovascular death), as expected, since few of the inputs bear on tumours. The cause groups are those of the public file's underlying-cause recode, which NCHS perturbs for some records.

5.6 Secondary estimand T2: death with diabetes or hypertension listed

The linked file marks deaths on whose certificate diabetes, or hypertension, appears anywhere. The outcome is death with the mention by year t , competing with death without it. In the cohort 971 deaths carry a diabetes mention and 1,325 a hypertension mention; one death has no multiple-cause data and is censored at its time. A mention is not a diagnosis, and it is under-reported.

Flagged deaths are few per fold, so predictions are not scored fold by fold: the out-of-fold predictions of one repeat are pooled (each of about 47,800 subjects appears once), scored as one set, and a 95% interval comes from resampling clusters (cycle, stratum, PSU) 1,000 times, on repeat 0. The pooled set has 237 (diabetes) and 304 (hypertension)

The full input set predicts every cause of death better

Time-dependent AUC at 10 years, cause-specific Cox models, 25 folds

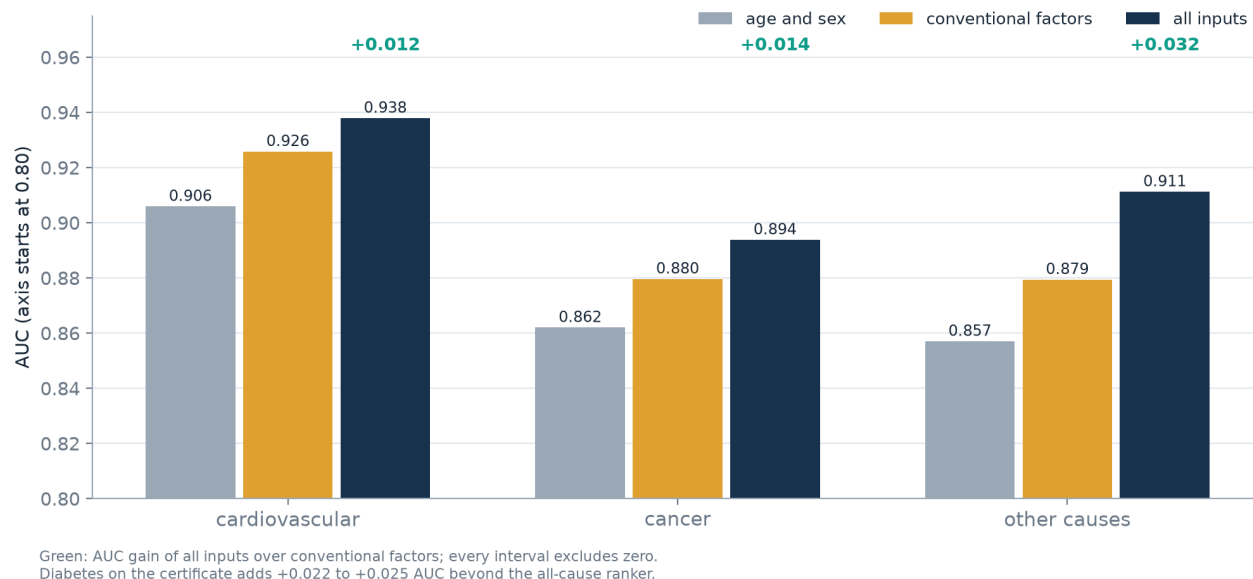


Figure 5: Time-dependent AUC at 10 years by cause of death (section 5.5).

flagged deaths by 5 years and 384 and 488 by 10 years, above the count of 100 the rule of section 4.5 requires before models are ranked. Two flag models are compared with the control that the rule names, the all-cause incidence of the full-input Cox model used as a ranker: a cause-specific Cox model for death with the mention, and the all-cause incidence times a penalised logistic estimate of the share of deaths that carry the mention (assuming the share does not depend on when death comes). Both use every input.

Outcome, horizon	All-cause ranker, AUC	Flag Cox, AUC minus ranker	Share model, AUC minus ranker
Diabetes listed, 5 y	0.948 [0.929, 0.964]	+0.0231 [+0.0118, +0.0368]	+0.0246 [+0.0150, +0.0360]
Diabetes listed, 10 y	0.943 [0.930, 0.955]	+0.0220 [+0.0129, +0.0325]	+0.0253 [+0.0175, +0.0339]
Hypertension listed, 5 y	0.926 [0.904, 0.945]	-0.0008 [-0.0079, +0.0055]	+0.0012 [-0.0033, +0.0062]
Hypertension listed, 10 y	0.935 [0.919, 0.949]	-0.0018 [-0.0069, +0.0037]	+0.0014 [-0.0023, +0.0052]

By the rule, the diabetes models add something to knowing who dies: both beat the ranker at both horizons with intervals above zero. With the conventional inputs only, the Cox model's gain at 10 years does not exclude zero (+0.0116 [-0.0003, +0.0238]). The hypertension models add nothing the all-cause ranker does not already have: the differences are within about 0.002 of zero with intervals that span it. Among deaths with multiple-cause data (7,102 in the pooled set), the implied share carrying the mention separates those with a diabetes mention from those without with an AUROC of 0.83 (Cox) to 0.84 (share model), and those with a hypertension mention with 0.61 for both. Calibration of the diabetes models is close to exact (observed over expected 0.95 to 1.01); for hypertension the Cox model is at 0.94 to 0.95 and the share model over-predicts by about 15% (0.83 to 0.86). The index of prediction accuracy is modest (0.095 and 0.14 for diabetes at 5 and 10 years; 0.043 and 0.065 for hypertension): the high AUCs come largely from age.

Results on repeats 1 to 4 (point estimates only; the repeats pool the same subjects, so they show stability across training splits, not independent confirmation) are within 0.003 in AUC of repeat 0 for every model and horizon. We

do not know why the hypertension mention is not predictable from the examination beyond the risk of dying; one possible reading, which we did not test, is that it is listed on certificates for reasons tied to the death more than to the person’s blood pressure.

5.7 Secondary estimand T3: expected time lived and mortality-equivalent age

A predicted curve gives each subject an expected time lived in the next τ years, the restricted mean survival time (RMST). Calibration compares the mean prediction in ten equal-weight risk groups with the Kaplan-Meier RMST observed there, on the pooled out-of-fold predictions of repeat 0, on the cycles whose follow-up reaches τ (23,097 subjects and 3,266 deaths at 10 years; 8,976 and 1,905 at 15 years, which only the 1999 to 2002 cycles reach).

Model	Mean absolute gap, 10 y (years)	Slope, 10 y	Uno C of the RMST, 10 y	Mean absolute gap, 15 y	Slope, 15 y
age, sex (Cox)	0.045	0.95	0.855	0.094	0.96
standard	0.036	1.01	0.876	0.087	1.01
(cause-specific Cox)					
all	0.035	1.02	0.898	0.058	1.01
(cause-specific Cox)					
spline Cox, all	0.025	1.01	0.900	0.089	1.00

By the rule of section 4.5 (a gap of at most 0.25 years and a slope in $[0.9, 1.1]$) every model is calibrated at both horizons, by a wide margin on the gap. The mean predicted and observed 10-year RMST are 9.51 and 9.53 years for the best models. Repeats 1 and 2 give the same figures within 0.005 years of gap. This is not a strong test of the models: the 10-year RMST is within 0.1 years of its ceiling for five of ten risk groups, where nothing can be wrong by much, and the discriminating information sits in the first two groups (predicted 6.9 and 9.1 years for the full-input model, observed 6.9 and 9.2).

Mortality-equivalent age. For each subject, the age at which a Gompertz life table gives the same expected time over 10 years. The table is fitted per sex to the training subjects of the subject’s fold by weighted maximum likelihood with delayed entry, so nothing about a test subject touches it. Ages are clamped to 18 to 85 and the clamps counted: 6% of subjects (1,356 of 23,097 for the full-input Cox model) fall outside the table’s range. They sit at both ends: 5% to 7% of subjects under 30 are healthier than any age in the range can express, so their equivalent age is the lower bound, and 41% of those aged 80 and over (10% of the seventies) have an expectation below that of an 85-year-old, so theirs is the upper bound. For these subjects the equivalent age is a bound and not a measurement, and the acceleration of the oldest in particular is censored from above; the hazard ratios below mix measured and bounded values and should be read with that in mind.

The association of the acceleration (equivalent age minus age) with outcomes over the whole follow-up, by Cox models adjusted for age and sex, with survey weights and variance clustered on PSUs:

Outcome (events)	Full-input Cox	Spline Cox	Age and sex only (control)
All-cause death (5,220)	1.72 [1.68, 1.77]	1.67 [1.63, 1.72]	0.84 [0.61, 1.15]
Cardiovascular (1,664)	1.76 [1.68, 1.84]	1.73 [1.65, 1.81]	0.55 [0.31, 0.97]
Cancer (1,114)	1.54 [1.46, 1.62]	1.50 [1.43, 1.58]	1.71 [1.11, 2.63]
Other (2,442)	1.80 [1.73, 1.87]	1.74 [1.68, 1.80]	0.75 [0.47, 1.20]
Diabetes mention (617)	2.04 [1.89, 2.19]	1.98 [1.85, 2.13]	0.38 [0.17, 0.86]
Hypertension mention (842)	1.66 [1.56, 1.77]	1.63 [1.53, 1.73]	0.34 [0.15, 0.77]

Hazard ratios per 5 years of acceleration. They are large and, as the rule of reading fixed in advance says, partly by construction: the acceleration is a function of a model fitted to predict death, evaluated on the same outcome. The control shows what is and is not construction. The same quantity from a model that sees only age and sex is a function of age and sex, which the Cox model adjusts for, and has no association with death (0.84, interval spanning

one); the intervals of its other rows are wide, since little of it remains after adjustment, and are not read. What is informative is the contrast across outcomes for the full-input models: the acceleration tracks diabetes-mentioned deaths most closely (about 2.0) and cancer deaths least (about 1.5), consistent with the cause-specific accuracy of section 5.5. Nothing here shows that the acceleration measures biological ageing, or that changing it would change a risk.

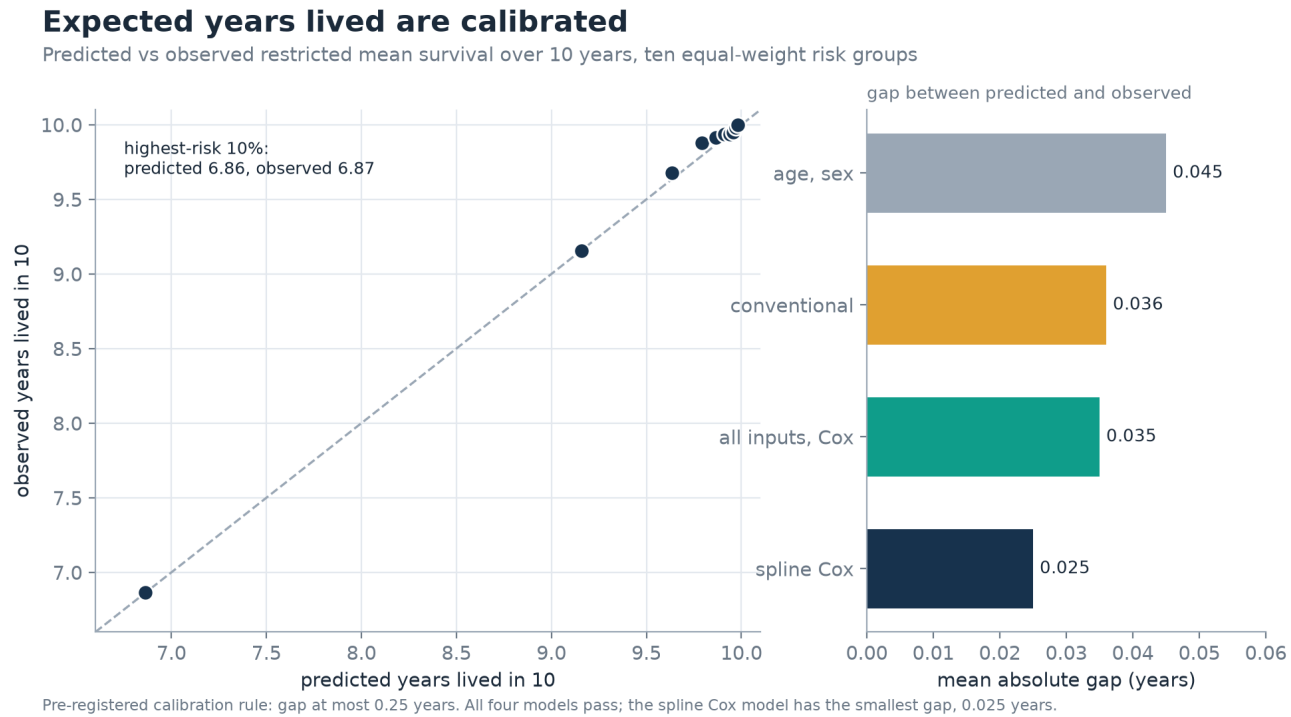


Figure 6: Calibration of expected time lived (section 5.7).

5.8 Secondary estimand T4: prevalent conditions at the examination

Eight conditions and one variant (diabetes defined with fasting glucose, on the fasting subsample and its weights) are labelled from the questionnaire and laboratory files (Appendix C). These are classifications at the same visit, not forecasts: a label defined by a measurement is not predicted from that measurement. Models therefore see one of two input sets, reported side by side: **nondef** (demographics, body measures, recalled weights, smoking, alcohol and eating times; the same for every label) and **full** (every input except those that define the label or a drug that treats it). The reference is a logistic model of age and sex. Models are logistic regression with a penalty chosen by cross-validation (**logit**) and gradient-boosted trees (**hgb**), fitted unweighted on the training subjects whose label is known and scored on the test folds with the survey weights, over the same 25 folds. A guard specifies that no excluded column reaches a model’s design matrix; writing it found that the display names of a column subset were mislabelled, a defect of the printing only (the matrix was right), repaired.

Label (weighted prevalence)	Age, sex	nondef , trees	gain over age, sex	full , trees	gain over age, sex
Diabetes (0.114)	0.742	0.843	+0.102 [+0.094, +0.110]	0.895	+0.153 [+0.146, +0.160]
Diabetes, fasting subsample (0.124)	0.745	0.834	+0.089 [+0.076, +0.102]	0.879	+0.134 [+0.125, +0.144]
Hypertension (0.371)	0.782	0.820	+0.037 [+0.034, +0.041]	0.840	+0.058 [+0.054, +0.062]

Label (weighted prevalence)	Age, sex	nondef , trees	gain over age, sex	full , trees	gain over age, sex
Kidney markers (0.151)	0.749	0.787	+0.039 [+0.032, +0.045]	0.845	+0.096 [+0.086, +0.107]
Anaemia (0.063)	0.661	0.791	+0.130 [+0.118, +0.143]	0.919	+0.259 [+0.245, +0.273]
High cholesterol (0.398)	0.731	0.751	+0.020 [+0.016, +0.023]	0.790	+0.059 [+0.053, +0.065]
Osteoporosis, age 50 and over (0.060)	0.848	0.852	+0.004 [-0.002, +0.011]	0.870	+0.022 [+0.014, +0.030]
Depression, PHQ-9 of 10 or more (0.076)	0.563	0.733	+0.170 [+0.147, +0.194]	0.819	+0.256 [+0.234, +0.278]
Sleep trouble told to a doctor (0.270)	0.603	0.672	+0.068 [+0.060, +0.076]	0.787	+0.184 [+0.173, +0.194]

AUROC, survey-weighted, mean over 25 folds; the gain carries the corrected resampled t-test interval. By the rule of section 4.5 (a gain of at least 0.02 with an interval excluding zero) the **full** models are useful for all nine labels, and the **nondef** models for seven: not osteoporosis (gain 0.004), and not high cholesterol, whose gain of 0.0197 misses the threshold by 0.0003. The logistic models give almost the same figures as the trees (full: within 0.01 for most labels; the trees add 0.011 to 0.015 for anaemia, kidney markers and high cholesterol). Calibration slopes lie between 0.94 and 1.21. The conventional risk-factor logistic model, restricted to what does not define the label, sits between age-sex and **nondef** for most labels and above **nondef** for kidney markers and high cholesterol.

How to read it. The **full** figures include correlates measured on the same blood draw as the label: the red-cell and white-cell indices for anaemia, the liver enzymes and uric acid for diabetes. They show what the examination contains, not what a person’s history predicts. The **nondef** figures are the more interesting: from age, body size, weight history, smoking, alcohol, income and eating times, diabetes is ranked with an AUROC of 0.84, depression with 0.73 and anaemia with 0.79. Labels are complete-case: a subject with an unknown label is dropped, not counted negative, and the weights are the examination weights, which are not exact for labels from the laboratory subsamples (urine albumin in particular). Bone density was not used for osteoporosis, whose label is the recalled diagnosis in those aged 50 and over, because its reference values could not be verified against a source.

5.9 Secondary estimand T5: undiagnosed disease

Among subjects who do not report the diagnosis (and, for diabetes and hypertension, are not treated), the label is the measured criterion: HbA1c of 6.5% or more, blood pressure of 140/90 or more, an eGFR below 60 or an albumin-to-creatinine ratio of 30 mg/g or more, total cholesterol of 240 mg/dL or more. It is one measurement, not a confirmed diagnosis, and “undiagnosed” rests on recall. A screen uses the **nondef** inputs only, since a screening tool cannot use the laboratory value it replaces, and a call is a score at or above the threshold that gives 90% specificity among the training subjects, estimated on cross-fitted predictions so that a flexible model’s training fit does not flatter it. The comparator is a model of age and body-mass index. The rule of section 4.5 claims value only if sensitivity at that threshold beats the comparator with a cluster-bootstrap interval excluding zero. Folds of repeat 0 are pooled; predictive values are restated at the weighted prevalence of the screened population.

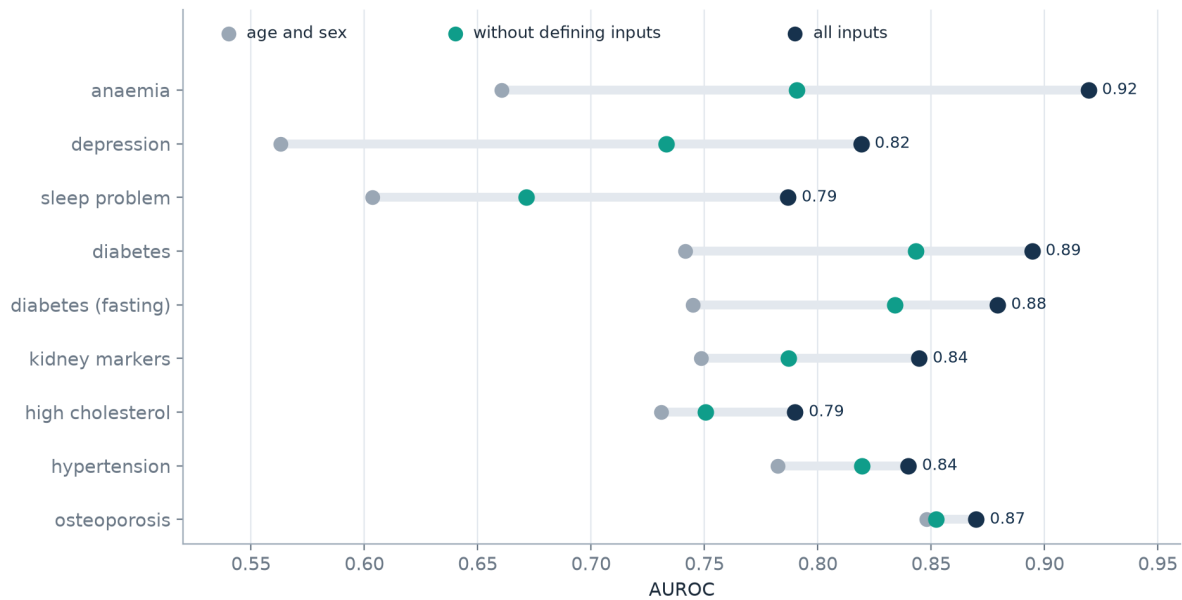
Undiagnosed (screened, cases, prevalence)	Screen	Sensitivity	Specificity	Positive predictive value	Minus age, BMI at the training threshold [95% CI]	Minus age, BMI at matched 90% specificity [95% CI]
Diabetes (39,837; 1,127; 2.0%)	age, BMI	0.332	0.915	0.073	comparator	comparator

Undiagnosed (screened, cases, prevalence)	Screen	Sensitivity	Specificity	Positive predictive value	Minus age, BMI at the training threshold [95% CI]	Minus age, BMI at matched 90% specificity [95% CI]
	nondef logistic	0.305	0.943	0.099	-0.027 [-0.064, +0.013]	+0.072 [+0.031, +0.114]
	nondef trees	0.330	0.933	0.091	-0.002 [-0.047, +0.039]	+0.048 [+0.000, +0.092]
	age, BMI	0.337	0.922	0.308	comparator	comparator
Hypertension (30,467; 3,325; 9.4%)	nondef logistic	0.326	0.938	0.352	-0.011 [-0.029, +0.007]	+0.035 [+0.011, +0.053]
	nondef trees	0.308	0.937	0.335	-0.028 [-0.046, -0.012]	+0.018 [-0.004, +0.037]
	age, BMI	0.369	0.928	0.422	comparator	comparator
Kidney markers (32,408; 5,188; 12.4%)	nondef logistic	0.350	0.940	0.454	-0.019 [-0.030, -0.008]	+0.021 [+0.008, +0.033]
	nondef trees	0.359	0.940	0.458	-0.010 [-0.021, +0.001]	+0.022 [+0.007, +0.035]
	age, BMI	0.077	0.937	0.116	comparator	comparator
High cholesterol (27,695; 2,901; 9.8%)	nondef logistic	0.171	0.914	0.176	+0.094 [+0.074, +0.116]	+0.081 [+0.056, +0.105]
	nondef trees	0.223	0.902	0.198	+0.146 [+0.123, +0.170]	+0.106 [+0.076, +0.130]

By the rule as written, only high cholesterol is claimed: the **nondef** models beat age and body-mass index at the training threshold with intervals above zero, and no screen beats the comparator for diabetes, hypertension or kidney markers. That reading is confounded: the training-derived thresholds gave test specificities of 0.90 to 0.94, not 0.90 (they were chosen on unweighted cross-fitted training predictions and applied to weighted test data), and a model that is more specific is less sensitive by construction. The final column re-reads every screen at exactly 90% specificity on the pooled set (a threshold no one could set in advance; **post hoc, not pre-registered**). There the **nondef** models are ahead of age and body-mass index for all four conditions by 2 to 11 points of sensitivity (the trees for hypertension do not exclude zero). Both readings agree on the size of the claim: sensitivity at 90% specificity is 0.41 to 0.45 for diabetes, hypertension and kidney markers and about 0.2 for cholesterol, and at the real prevalence of 2% for diabetes about nine of ten positive calls are false. For cholesterol, sex alone accounts for a part of the gain over age and body-mass index (age-sex gains 0.035 at the training threshold). We claim no clinical screening value for any of these.

What the examination reveals about the present state

AUROC for conditions already present: age and sex, without the inputs that define the label, and with all inputs

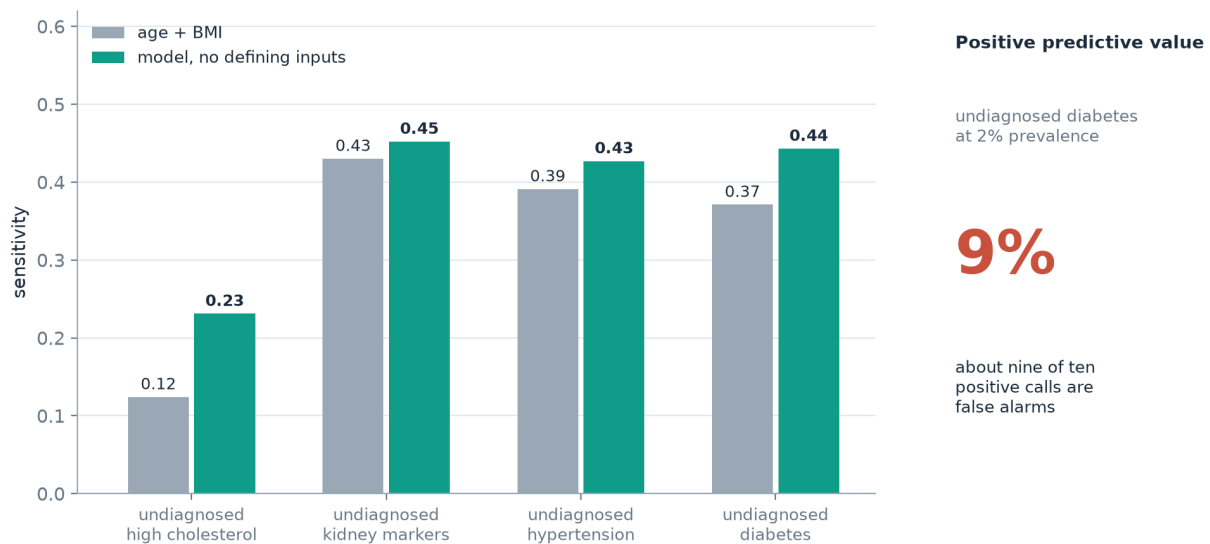


Gradient-boosted classifier, 25 folds. 'All inputs' includes correlates measured at the same visit, so it reads as detection, not prediction.

Figure 7: Prevalent conditions (section 5.8).

Screening for undiagnosed disease is still weak

Sensitivity at 90% specificity: age and BMI alone vs the best non-defining model



The model beats age+BMI at matched specificity for all four (a post hoc reading); the pre-registered rule credits only cholesterol.

Figure 8: Screening for undiagnosed disease (section 5.9).

5.10 Accelerometer summaries as an input block

The rule is in section 4.6. Of the 10,332 examinees of the 2003-2004 and 2005-2006 cycles with at least four valid days of recording, 6,952 are in the cohort (adults eligible for linkage); 5,877 of them are outside the locked test, and 806 died within ten years. The summaries have plausible levels: the median adult wears the monitor 14 hours a day, averages 285 counts per minute and spends 15 minutes a day at or above 2,020 counts. An early version of the extraction called every day of every subject valid with 24 hours of wear, because the transport file stores a zero count as 5.4e-79 and no zero minute was ever seen; the specification for the extraction now includes that value.

Metric (10 years unless stated)	Without the block	With the block	Paired difference [95% CI, corrected]
Brier score	0.05186	0.05097	-0.00089 [-0.00167, -0.00011]
Uno concordance	0.9028	0.9034	+0.0006 [-0.0016, +0.0028]
Uno concordance, 5 years	0.9100	0.9123	+0.0022 [-0.0017, +0.0061]
Calibration slope	1.112	1.104	-0.008 [-0.044, +0.028]

By the rule of section 4.6 the block is useful: the interval of the Brier difference lies below zero. The evidence is thin and the effect small. The upper end of the interval is within 0.0001 of zero, the gain is under 2% of the Brier score, and discrimination does not resolvably change. The comparison is on 5,877 subjects, so both arms are trained on about 4,700 and are worse estimated than the models of the other sections (the integrated Brier score over 15 years cannot be computed: no cycle here is followed that long). One reading, which we did not test, is that what a hip accelerometer says about the next ten years is largely already in the examination, in age, body size, blood tests and self-reported health, with a small remainder. A sequence model on the minute counts, which the roadmap asked for, was not tried.

5.11 The residual network: is there structure beyond the additive model?

Design and rule: section 4.7, fixed before the first run. An earlier plan for this experiment, written the day before and kept outside the repository (a larger design with piecewise-linear embeddings and a sweep of sixty configurations), was never run; the agent's two attempts at it (section 5.1) produced a candidate that crashed on the real folds. The design here is the smaller one of section 4.7, and the trained networks are the first that exist.

Capacity sweep, repeat 0 (5 folds), network minus the spline Cox model it is built on (negative is better):

Width, hidden layers	Integrated Brier, 0 to 15 y	Brier, 10 y
16, 1	-0.00006 [-0.00045, +0.00033]	-0.00024 [-0.00071, +0.00023]
64, 1	-0.00014 [-0.00062, +0.00034]	-0.00028 [-0.00079, +0.00023]
256, 1	-0.00011 [-0.00062, +0.00040]	-0.00024 [-0.00075, +0.00026]
64, 2	-0.00033 [-0.00110, +0.00045]	-0.00048 [-0.00121, +0.00024]
64, 3	-0.00029 [-0.00091, +0.00033]	-0.00034 [-0.00108, +0.00040]

Every configuration is nominally ahead of the spline model and none is resolved. The headline configuration by the rule fixed in advance, the largest gain in integrated Brier score, is width 64 with two hidden layers.

Confirmation, repeats 1 to 4 (20 folds), headline configuration against the spline Cox model:

Metric	Spline Cox	Residual network	Paired difference [95% CI, corrected]
Integrated Brier, 0 to 15 y	0.04599	0.04591	-0.00008 [-0.00030, +0.00015]
Brier, 10 y	0.05344	0.05327	-0.00017 [-0.00037, +0.00003]

Metric	Spline Cox	Residual network	Paired difference [95% CI, corrected]
Uno concordance, 10 y	0.90078	0.90130	+0.00052 [-0.00049, +0.00152]

By the rule of section 4.3 the residual network is not useful: the integrated Brier score improves by 0.00008, a tenth of the 0.001 the rule requires, with an interval spanning zero. The pooled analysis with a cluster bootstrap (repeats 1 and 2) gives -0.00011 [-0.00026, +0.00004] and -0.00009 [-0.00023, +0.00005] in the integrated score, and, at ten years, -0.00015 [-0.00030, -0.00000] and -0.00024 [-0.00038, -0.00009]: a gain at ten years that is resolved there and is 0.3% to 0.5% of the score. The networks stopped after a median of 34 epochs, and in 3 of 25 folds at epoch zero, which is the spline model.

Feature blocks, confirmation folds, the headline network given one block of inputs at a time (the spline base unchanged):

Block	Integrated Brier [95% CI]	Brier, 10 y [95% CI]
all inputs	-0.00008 [-0.00030, +0.00015]	-0.00017 [-0.00037, +0.00003]
examination measurements	-0.00007 [-0.00027, +0.00013]	-0.00016 [-0.00036, +0.00004]
eating-time features	+0.00002 [-0.00004, +0.00008]	+0.00001 [-0.00007, +0.00008]
recalled diagnosis ages and weights	-0.00004 [-0.00015, +0.00007]	-0.00006 [-0.00018, +0.00005]
questionnaire answers	-0.00000 [-0.00022, +0.00022]	-0.00001 [-0.00016, +0.00014]

What gain there is, is carried by the examination measurements: the network on that block alone reproduces almost all of the all-input gain, and the eating-time features, the history and the questionnaire answers add nothing resolvable. The shuffled-outcome control (trained on outcomes permuted among the training subjects) gives +0.00002 [-0.00003, +0.00007] and stopped at epoch zero in 6 of 20 folds, with a median of 6 epochs elsewhere: the training procedure does not manufacture a gain.

The answer to the paper’s question, for this data size and this recipe, is no: a small network trained on the residual of the spline Cox model does not find structure worth a thousandth of a Brier point in the integrated score, and the gain it may have at ten years, about 0.0002, is a fifth of what the rule requires. This is a result for a multilayer perceptron on standardised inputs with one set of optimiser settings, 38,000 training subjects and about 5,700 deaths; it is not a proof that no structure exists, and a larger sample would be the way to test it.

5.12 The accelerometer series against the summaries

Rule: section 4.8. Four arms on the restricted folds of section 5.10 (5,877 subjects outside the locked test, 25 folds): the base (spline Cox, all inputs), the base with the eight summaries, the base with a convolutional network on the ten-minute grid of the recording, and the base with both. The series network stopped after a median of 27 epochs (never at epoch zero); with the summaries already in the base it stopped after 7 and in 8 of 25 folds at epoch zero.

Comparison	Brier, 10 y: difference [95% CI, corrected]	Uno C, 10 y: difference [95% CI]
series network minus base	-0.00079 [-0.00130, -0.00028]	+0.0014 [-0.0004, +0.0032]
series network minus summaries	+0.00010 [-0.00052, +0.00072]	+0.0008 [-0.0018, +0.0033]
summaries and series minus summaries	+0.00001 [-0.00025, +0.00027]	+0.0003 [-0.0007, +0.0013]
summaries and series minus base	-0.00088 [-0.00172, -0.00004]	+0.0009 [-0.0016, +0.0035]

By the rule the series network is useful beyond the base: the interval of the Brier difference lies below zero. It is not better than the summaries: the interval against the base with summaries spans zero, with a point estimate that is slightly worse (+0.0001) and a gain over the base (-0.0008) about the size of the summaries’ (-0.0009, section 5.10). Putting both together adds nothing to either. Whatever a hip accelerometer says about ten-year mortality

beyond the examination, in this sample of a few thousand adults, the eight summaries carry as much of it as a convolutional network on the minute pattern does, and no more can be resolved. The comparison has little power (the intervals are 0.0006 to 0.0010 wide), so it excludes a large advantage of the series, not a small one.

What the pre-registered tests resolved, and what they did not

Difference in Brier score (negative = better). 95% intervals; dashed line = the 0.001 gain the rule required

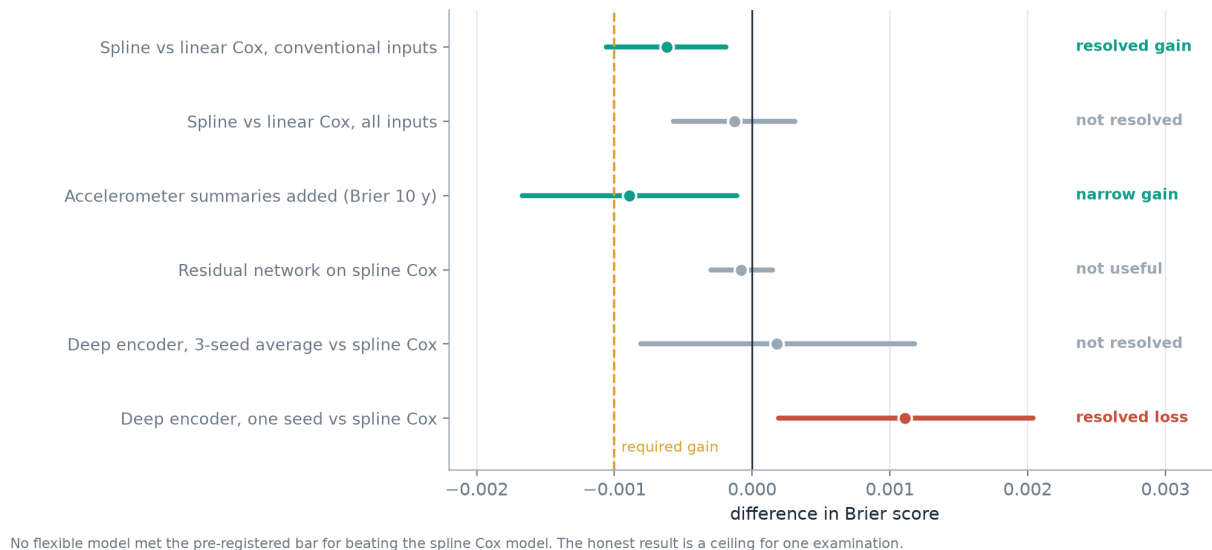


Figure 9: Which pre-registered comparisons were resolved (sections 5.2, 5.10, 5.11).

5.13 Which inputs the model relies on

The rules are those of section 4.9; the grouped analysis was added after the single-input result was seen and is post hoc. Repeat 0, five folds, about 9,560 test subjects and 1,420 deaths per fold, ten shuffles per input (`baselines/drivers.py`; the per-input numbers are identical with and without the grouped analysis). Of the 67 inputs, 38 carry independent information by the rule (mean fall at least 0.001 nats per event and positive in all five folds); 29 do not. The 22 largest:

rank	input	fall in partial log-likelihood [SE]	without deaths in the first 2 y	hazard ratio
1	age	1.3713 [0.0552]	1.4378	p90 vs p10: 26.38
2	sex	0.0345 [0.0036]	0.0288	male vs female: 1.44
3	race_ethnicity	0.0314 [0.0012]	0.0341	mexican_american vs nh_white: 0.66; nh_black vs nh_white: 0.79; other vs nh_white: 0.69; other_hispanic vs nh_white: 0.65
4	smoking	0.0314 [0.0040]	0.0315	current vs never: 1.47; former vs never: 1.08
5	alt_ul	0.0310 [0.0045]	0.0324	p90 vs p10: 0.60
6	urine_albumin_ugml	0.0263 [0.0023]	0.0231	p90 vs p10: 1.44
7	rdw_pct	0.0235 [0.0042]	0.0255	p90 vs p10: 1.50
8	weight_kg	0.0213 [0.0046]	0.0116	p90 vs p10: 0.66
9	self_rated_health	0.0182 [0.0008]	0.0178	p90 vs p10: 1.29

rank	input	fall in partial log-likelihood [SE]	without deaths in the first 2 y	hazard ratio
10	waist_cm	0.0178 [0.0040]	0.0154	p90 vs p10: 1.31
11	ggt_ul	0.0174 [0.0007]	0.0164	p90 vs p10: 1.27
12	ast_ul	0.0166 [0.0021]	0.0174	p90 vs p10: 1.30
13	prescriptions	0.0143 [0.0009]	0.0129	p90 vs p10: 1.36
14	mcv_fl	0.0126 [0.0026]	0.0110	p90 vs p10: 1.31
15	albumin_gdl	0.0117 [0.0026]	0.0074	p90 vs p10: 0.75
16	income_poverty_ratio	0.0106 [0.0015]	0.0095	p90 vs p10: 0.77
17	marital	0.0097 [0.0006]	0.0088	divorced vs married: 1.17; never_married vs married: 1.28; partner vs married: 1.21; separated vs married: 1.17; widowed vs married: 1.17
18	lymphocyte_pct	0.0075 [0.0013]	0.0078	p90 vs p10: 0.84
19	urine_creatinine_mgdl	0.0070 [0.0015]	0.0086	none given
20	bmi	0.0069 [0.0023]	0.0063	p90 vs p10: 0.85
21	phq9	0.0060 [0.0006]	0.0063	none given
22	told_stroke	0.0059 [0.0019]	0.0029	yes vs no: 1.31

Age dominates everything: shuffling it costs 1.37 nats per event, about forty times the next single input (sex, 0.035), and the model’s hazard ratio between the 90th and 10th percentile of age is 26 (range over folds 21 to 37). Because related inputs share credit when one is shuffled alone (body weight, waist and body-mass index; glucose, glycated haemoglobin and the diabetes question), the groups are the fairer reading:

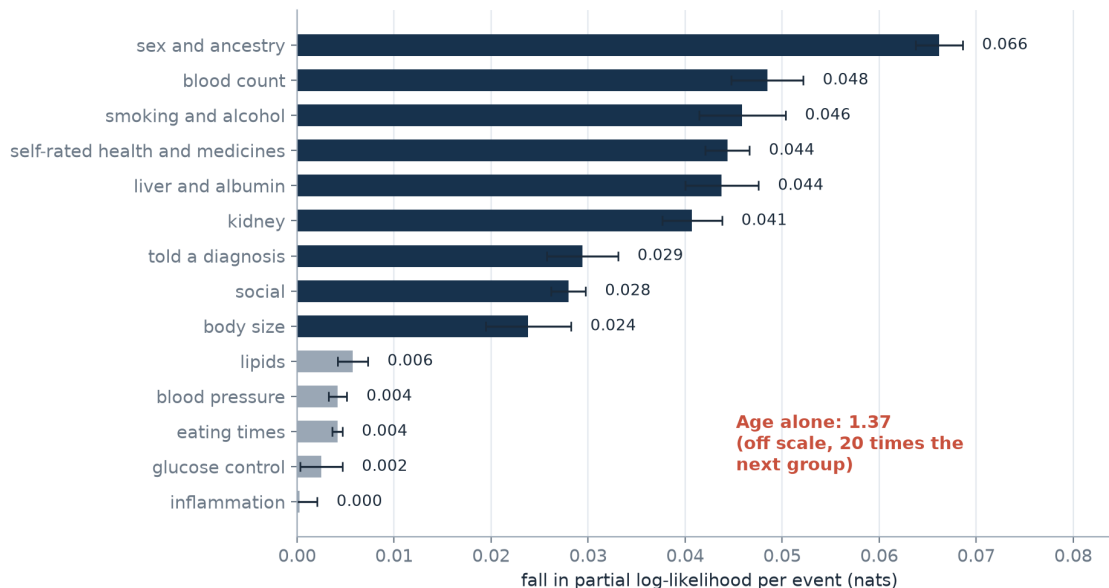
group	fall in partial log-likelihood [SE]	folds positive
age	1.3665 [0.0603]	5/5
sex and ancestry	0.0662 [0.0024]	5/5
blood count	0.0485 [0.0037]	5/5
smoking and alcohol	0.0459 [0.0045]	5/5
self-rated health and medicines	0.0443 [0.0023]	5/5
liver and albumin	0.0438 [0.0038]	5/5
kidney	0.0407 [0.0031]	5/5
told a diagnosis	0.0294 [0.0037]	5/5
social	0.0280 [0.0018]	5/5
body size	0.0238 [0.0044]	5/5
lipids	0.0057 [0.0016]	5/5
blood pressure	0.0042 [0.0010]	5/5
eating times	0.0042 [0.0005]	5/5
glucose control	0.0025 [0.0022]	4/5
inflammation	0.0003 [0.0018]	4/5

Beyond age, no group stands alone: sex and ancestry, the blood count, smoking and alcohol, self-rated health with the number of prescriptions, the liver enzymes with albumin, and the kidney markers each cost 0.04 to 0.07 nats. The conventional cardiovascular measurements cost little once these are present: lipids 0.006, blood pressure 0.004, and glucose control 0.002 (positive in four folds of five), which is consistent with the conventional-factor score trailing the full inputs in section 5.2.

Direction, as the hazard ratio between the 90th and 10th percentile or between levels. Higher hazard goes with a wider red-cell distribution width (1.50), current smoking against never smoking (1.47; former smoking 1.08), albumin in the urine (1.44), male sex (1.44), more prescriptions (1.36), a larger waist (1.31), a history of stroke (1.31), poor

Which groups of measurements the model relies on

Fall in held-out partial log-likelihood when a group is shuffled among test subjects, 5 folds, bars show mean and SE



Post hoc grouping of related inputs. Correlated inputs share credit when shuffled one at a time, so groups are the fairer reading.

Figure 10: Groups of inputs by how much the model loses when they are shuffled.

self-rated health (1.29), heart failure (1.27) and cancer (1.23). Lower hazard goes with higher serum albumin (0.75), higher income relative to the poverty line (0.77), a higher lymphocyte share (0.84), a higher body-mass index (0.85) and, strikingly, a higher ALT (0.60) and a heavier body weight (0.66).

The last two are the warning against reading these as advice. A low ALT and a low weight at a fixed waist and body-mass index are markers of frailty and of low muscle mass, and the model uses them for that. The ratios are conditional on every other input, set one at a time over the test subjects, so weight, height, waist and body-mass index cannot be read separately.

Early deaths. Without the subjects who die in the first two years (about 9,360 test subjects and 1,220 deaths per fold), the rank correlation of the 67 importances is 0.92 and eight of the ten most important inputs are the same. Smoking, red-cell distribution width, albumin in the urine and self-rated health keep their importance and direction. The importance of body weight falls from 0.021 to 0.012 and its ratio moves from 0.66 to 0.73, and that of a stroke history from 0.006 to 0.003: the weight and stroke associations are partly illness that was already present.

What this shows and does not show. It shows which measurements the fitted model uses for the ranking of who dies within the follow-up, with a sign, in a cohort of 47,816 adults and 7,103 deaths. It does not show that changing any of them would change anything: the one behavioural input with a large and consistent ratio is smoking, which other studies establish as causal and this one does not, and the other large ratios are measurements of disease, or of how ill a person is. The variables that lead to a long life in the sense a reader would want, those a person can act on, need the design of Appendix F.

6. Discussion

What the survival results license. On this cohort, with the information a single examination contains, a regularised Cox model with a spline in every continuous input is the best model we measured by point estimate, and it is not resolvably better than the same model without splines when all inputs are used. The deep set encoder is resolvably worse than the spline model by 0.0011 in the integrated Brier score by the corrected t-test over 25 folds. On the pooled out-of-fold predictions of repeat 0 with a cluster bootstrap the figure is +0.0007 [-0.0001, +0.0016] in the integrated Brier score, not resolved, and +0.0011 [+0.0003, +0.0018] at ten years, resolved: the

Direction and size of the association with death

Hazard ratio between the 90th and 10th percentile (or level vs reference); range over 5 folds; log scale

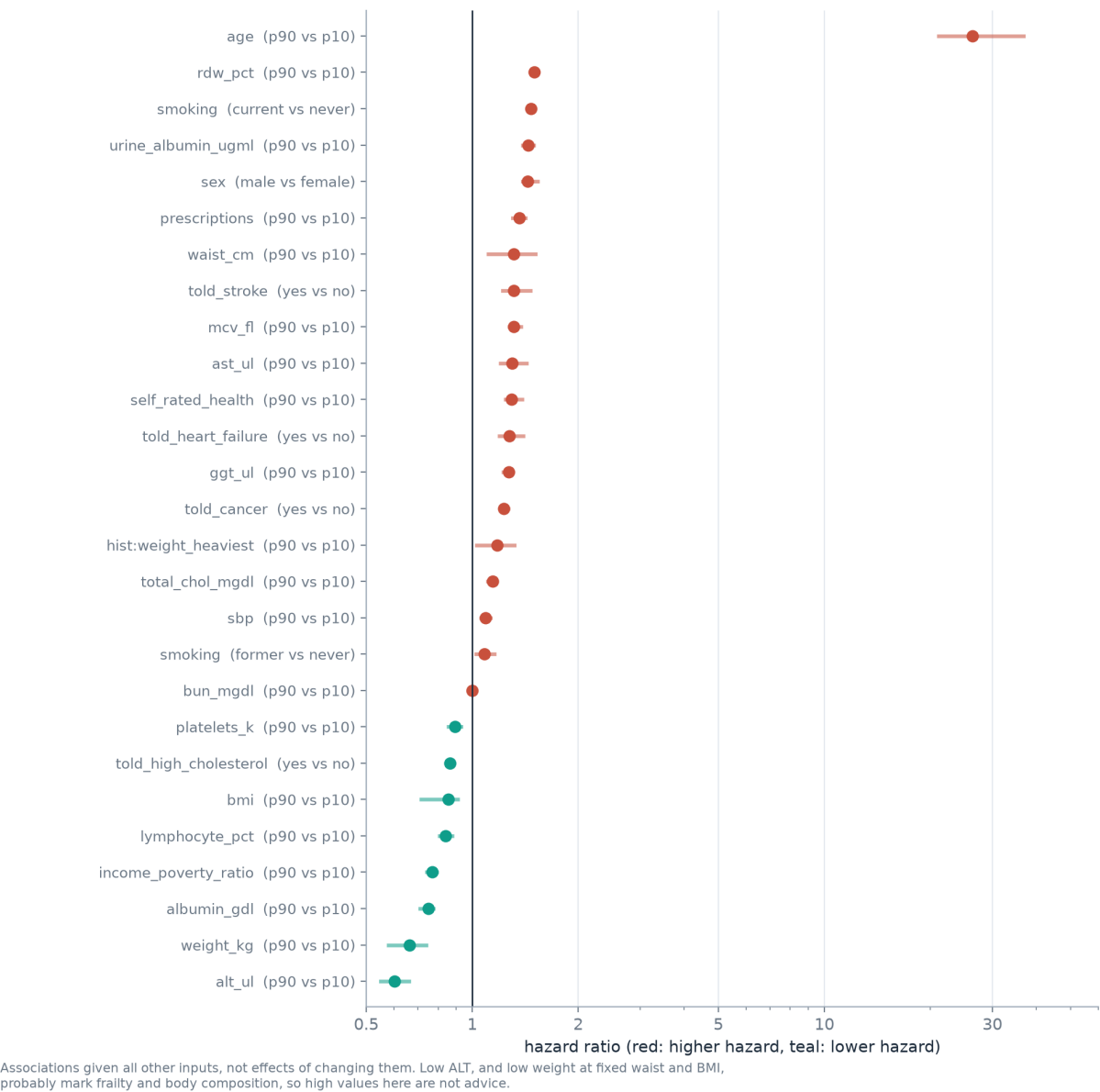


Figure 11: Hazard ratios of the informative inputs.

deep encoder is behind at ten years on both analyses and the integrated difference does not survive the second. Failing to find a gain is not proof that none exists: our power to resolve a true difference of 0.001 is limited (section 3 and the audit in Appendix B), and the residual-network experiment, the one that tests “is there structure the additive model misses”, was not delivered by the agent (section 5.1) and was run by the supervisor (section 5.11): a negative result for one network, one recipe and this sample size. What can be said is narrower and still useful: three differently built learners (a nonlinear Cox model, a logistic model with inverse-probability-of-censoring weights, and a piecewise- exponential additive model) land within 0.0005 of one another, and the deep encoder, whose curve is still falling with more data, has not caught up with them at this sample size. Whether it would with several times more subjects is a question for data that are not on this machine.

What the learning curves license. The Cox curves are flat from half of the training subjects onward, so more of the same data will not improve them; the learners with more flexible hazards are still improving. Both statements come from fold means of a metric whose folds are positively correlated, with one training seed for the deep recipes; the exponents of the flexible learners are determined and the asymptotes of every fit are extrapolations, hypotheses for a larger sample.

What the agent results license. Of four substantive tasks given to the loop (a nonlinear Cox model, its specification tests, a learning-curve fitting tool, a residual network), two were accepted (the second after infrastructure repairs and one retry) and two were not delivered. The accepted work was checked by the supervisor against hidden checks written before the run and then read against the task; reading found two gaps (a missing specification test, and a documentation item that my own check had failed to test). The failures were informative: the 27-billion-parameter model decoded greedily loops on one action; with reasoning enabled it spends its output budget before writing; with a small server context a long conversation is refused; and a larger multi-part task (a neural network with a fitted likelihood and a learned multiplier) exceeded what it completed in three attempts. Each infrastructure defect was repaired in the layer that owned it, with a regression specification, and a defect of the supervisor’s own checks was found by the same process (F-017, F-019). We report the delivered fraction (2 of 4 tasks; 2 of the 10 attempts that reached a verdict in the runs after the repairs were accepted, the other 8 were rejected by a check or ran out of a limit) as a property of this model on these tasks, not as a general rate.

What the secondary estimands license. One examination predicts more than whether a person dies. For death by cause, adding the full set of inputs to the conventional risk factors improves the cause-specific Brier score for cardiovascular, cancer and other deaths (section 5.5), most for the residual group. For death with a diabetes mention on the certificate it adds to knowing who dies (section 5.6), and for a hypertension mention it adds nothing. Expected time lived is well calibrated by every model (section 5.7), and the mortality-equivalent age built on it tracks diabetes-mentioned and cardiovascular deaths more closely than cancer deaths, with a control that separates what is construction from what is information. Conditions at the examination are ranked well from the questionnaire and measurements alone (section 5.8), and the screens for undiagnosed disease built on non-laboratory inputs are weak: a sensitivity of 0.2 to 0.45 at 90% specificity, a positive predictive value of 9% for diabetes at its real prevalence (section 5.9). None of these is a forecast of incident disease: the data have no follow-up for it. What the examination shows is that one visit contains a calibrated mortality forecast by cause, an estimate of expected time lived, and classifications of conditions, some of whose labels and inputs come from the same blood draw, and that is the extent of it.

What the learning-and-promotion results license. Only that the path works and can say no: a candidate adapter was trained, reloaded and judged on held-out tasks under a rule fixed beforehand, and was rejected without disturbing the model in use. It licenses nothing about whether the loop’s learning improves the agent; that needs more accepted runs, a trainer that takes multi-step tool conversations, and an evaluation set with a baseline far from zero. The autonomy stage reached is *supervised local*: the loop works on local models with the supervisor’s diagnostic repairs recorded; no unaided local evaluation on unseen tasks was run.

7. Limitations

- One examination per person: the data cannot answer whether repeated measurements add information, which is the regime where sequence representations have been reported to help. The accelerometer minute data for two cycles (2003 to 2006) were reduced to eight daily summaries (section 5.10); a model of the minute sequences themselves, which would be a long-sequence experiment, was not tried.
- One country and era; mortality is the only outcome observed after the examination; follow-up is administratively censored, so the 15-year metric rests on the earliest cycles alone.

- Analysis-time reuse of the cross-validation folds: capacity and recipe choices for the deep encoder were made on repeat 0 and confirmed on others, but baselines, the spline model and the recipe sweep share the same partition; all configurations tried are reported and none was selected on a locked test.
- The pooled out-of-fold paired bootstrap that the protocol (section 4.2) prefers to the corrected resampled t-test was specified as a loop task and not run for the primary comparisons: every interval of sections 5.2 to 5.5, 5.7 (calibration), 5.8 and 5.10 is a corrected resampled t-test interval. Sections 5.6, 5.7 (associations) and 5.9 use a cluster bootstrap on pooled out-of-fold predictions of one repeat, which the other repeats do not confirm independently, since they pool the same subjects.
- The secondary estimands (sections 5.5 to 5.10) were written down before their numbers existed, but they are many: nine cause-horizon cells, two flags, eight conditions and four screens are reported, each against its own rule, and none is adjusted for the number of looks. A rule met by a narrow margin (the accelerometer block, the cholesterol gain of 0.0197) should be read as such.
- The matched-specificity reading of the screens is post hoc; the calibration summaries of section 5.5 come from one repeat; weights for labels from laboratory subsamples other than fasting glucose are the examination weights, which is not exact; the bone-density definition of osteoporosis was not used; creatinine is standardised by the correction each cycle's laboratory documentation states, and estimated filtration rate by an equation quoted from the literature, not verified against a second implementation.
- One input of the frozen timelines is defective: the question about weak or failing kidneys is asked under another name in the 1999-2000 cycle, so those subjects lack it. Refitting the spline Cox model without that input changes the integrated Brier score by -0.00001 [-0.00005, +0.00004] over 25 folds, so the timelines were left as they were frozen; the labels of section 5.9 read both names.
- The agent was evaluated on four tasks and on one local model; the learning experiment trained on nine accepted runs, seven of which are the same task. Nothing here shows general autonomy.
- The residual-network experiment and the calibration-curve additions (integrated calibration index, E50 and E90) were specified for the agent and not delivered by it; the supervisor wrote and ran them (sections 5.5 and 5.11). The residual network is one architecture, one set of optimiser settings and one training seed per fold, on 38,000 training subjects; the audit of the calibration code had found no defect in the censoring-aware quantities but that no summary of the curve was reported, which the new summaries address.

8. Reproducibility

Code revisions: brain b7997074, sven a6751d7, splinter at the commit that contains this file (`git log -1 -- papers/agent-driven-residual-survival/paper.md`). Data: the lifecourse build described in `samples/lifecourse/README.md` (timelines, partition and frozen criteria are content-addressed in `FROZEN.json`). Commands, in the order they were run:

```
# local model served by brain (CUDA); one process on the device at a time
BRAIN_BACKEND=cuda BRAIN_QWEN35_GGUF_CTX=98304 brain serve --openai 127.0.0.1:8788 --models-dir <models>
# a delegated task (sampling temperature recorded in the run's model_selected event)
BRAIN_API_KEY=<key printed by serve> agent-loop --temperature 0.7 run --workspace <clone> \
  --task-file <task.txt> --accept "behaviour=bash <accept.sh>" --accept-visible "harness-tests=..." \
  --protect <paths> --model "remote:brain/unsloth/Qwen3.8-27B-Q8_0" --allow-api-models \
  --max-output-tokens 40000 --max-attempts 3 --follow-ups 3 --json
# the spline baseline on the 25 folds, scored like every arm
python -I samples/lifecourse/baselines/run.py --data <data> --out <data>/baselines --baseline spline-cox-net-all
splinter-lifecourse external --data <data> --baseline spline-cox-net-all
splinter-lifecourse compare --data <data> --a external:spline-cox-net-all --b external:cox-net-all
# learning curves: shares of the training subjects, scored on the same folds
splinter-lifecourse recipe subsample --data <data> --share 0.25
python -I samples/lifecourse/baselines/run.py ... --train-ids <data>/recipe/_subsamples/p25-s1 --repeat 0
splinter-lifecourse recipe score --data <data> --name spline-cox-net-all-p25 --made-of "..."
# learning from accepted runs, and judging the candidate
agent-loop dataset --run <accepted run> ... --out train.jsonl
agent-loop train --dataset train.jsonl --base local:Qwen/Qwen3-8B --steps 30 --rank 16
samples/agent/loop/fixtures/eval-candidate.sh <out> <model-ref> anagrams:1 bisect:1 ...
# secondary estimands: the multiple-cause flags beside the timelines, then death by cause (T1)
splinter-lifecourse build --nhanes <nhanes> --mortality <mortality> --data <new dir> # timelines.jsonl i
```

```

python -I samples/lifecourse/baselines/run.py --data <data> --out <data>/baselines --baseline cs-cox-agesex
splinter-lifecourse causes --data <data> --model cs-cox-agesex --model cs-cox-standard --model cs-cox-all
# T2: death with a diabetes or hypertension mention
python -I samples/lifecourse/baselines/flags.py --data <data> --out <data>/baselines --model flag-cox-all
splinter-lifecourse flags --data <data> --ranker cs-cox-all --model flag-cox-all --model flag-share-all --
# the residual network (repeat 0 sweep, repeats 1-4 confirmation), its ablations and control, pooled comm
python -I samples/lifecourse/baselines/run.py --data <data> --out <data>/baselines --baseline resnet-cox-a
splinter-lifecourse compare --data <data> --a external:resnet-cox-all-w64-l2 --b external:spline-cox-net-a
splinter-lifecourse bootstrap --data <data> --a resnet-cox-all-w64-l2 --b spline-cox-net-all --repeat 1
splinter-lifecourse calibration --data <data> --model cs-cox-all --repeat 0
# the accelerometer: summaries, series grid, the four arms on the restricted folds
python -I samples/lifecourse/baselines/accel.py --zip <PAXRAW_C.zip> --zip <PAXRAW_D.zip> --out <data>/acc
python -I samples/lifecourse/baselines/accel_experiment.py --data <data> --out <data>/baselines --jobs 16
splinter-lifecourse external --data <data> --baseline accel-seq --subset
# learning curves on repeats 1-4 (CPU baselines, then the deep recipes under the GPU lock), then the analy
python -I samples/lifecourse/baselines/curves.py --data <data> --bootstrap 500
# T3: expected time lived, mortality-equivalent age, and its association with outcomes
splinter-lifecourse labels --nhanes <nhanes> --data <data>
python -I samples/lifecourse/baselines/conditions.py --data <data> --out <data>/conditions --jobs 16
splinter-lifecourse prevalence --data <data> --model agesex --model conv-logit --model nondef-logit --model
splinter-lifecourse screen --data <data> --model agebmi --model agesex --model nondef-logit --model nondef
splinter-lifecourse lifeexp --data <data> --model cs-cox-all --model spline-cox-net-all --repeat 0 --tau 1
python -I samples/lifecourse/baselines/assoc.py --data <data> --aa <aa dir>/aa-cs-cox-all-r0-t10.jsonl
# which inputs the survival model relies on (T7), and without deaths in the first two years
python -I samples/lifecourse/baselines/drivers.py --data <data> --out <dir> --jobs 5
python -I samples/lifecourse/baselines/drivers.py --data <data> --out <dir> --jobs 5 --exclude-early 2
# figures (matplotlib): python papers/agent-driven-residual-survival/figures.py img <dir>/drivers-all.json
agent-loop models judge --candidate <version> --baseline-results <dir> --candidate-results <dir>

```

Delegated tasks (task text and hidden acceptance scripts are kept outside the repository so that a worker’s checkout cannot contain them; their SHA-256 prefixes are given):

task	run	verdict	task / check digest
spline Cox model	run-20261007T203058.568-370c	accepted, attempt 2 of 3	76994035 / 54a7583d
its specification tests	run-20261007T205231.010-e99c	accepted, attempt 1 (README item unmet, F-017)	89fad691 / aac259b7
learning-curve tool	run-20261007T205818.780-7b26	not delivered (limits; cancelled at attempt 3)	2af4b8c9 / 9eb66011
residual network	run-20261007T215753.920-5a61	rejected after 3 attempts	4082373b / 4fe6cc75
residual network, continued, assisted by a hint	run-20261007T225456.244-8b2c	rejected after 2 attempts	4082373b / 4fe6cc75

Earlier delegated attempts of the first task ended before a model could be judged because of infrastructure defects (section 5.1): runs 6059, a64e (killed by the supervisor after its first two attempts looped), aa65, 9e1c and c44b.

Appendix A. Research log

Date	Milestone	Outcome
2026-10-07	M0 protocol and plan	Paper created; decision rule fixed (section 4.3)

Date	Milestone	Outcome
2026-10-07	M1 loop built	Section 4.4 describes the loop as implemented; specs pass with a scripted model; real-model result pending
2026-10-07	Reuse of the agent runtime	Event stream moved onto the runtime's hash-chained log; definitions in effect recorded through its discovery; two small additions to the runtime's SDK (chain , workspace), neither naming this project
2026-10-08	Learning from accepted runs	Trainer refused multi-step tool conversations (F-021); patch-form records trained an adapter; candidate rejected on eight held-out tasks (section 5.4)
2026-10-08	Secondary estimands pre-registered (section 4.5); T1 death by cause (section 5.5)	The full-input cause-specific Cox model beats the conventional risk-factor model for each of three causes at 10 years; calibration described, its bootstrap not run
2026-10-08	T2 death with a diabetes or hypertension mention (section 5.6)	Diabetes models beat the all-cause ranker (AUC +0.022 to +0.025, intervals above zero); hypertension models do not
2026-10-08	T3 expected time lived and mortality-equivalent age (section 5.7)	Every model calibrated by the section 4.5 rule; acceleration associates with cause and flag as described, and its age-sex control shows no association
2026-10-08	T4 prevalent conditions and T5 undiagnosed-disease screens (sections 5.8, 5.9)	Models beat age-sex for nine labels (full inputs) and seven (non-definitional); by the rule only cholesterol is a screen, at matched specificity all four are ahead of age and body-mass index (post hoc)
2026-10-08	Accelerometer summaries as an input block (sections 4.6, 5.10)	Rule fixed before the files were opened; the 10-year Brier score falls by 0.0009 with an interval just below zero, discrimination unchanged; an extraction defect (zero counts stored as 5.4e-79) found by checking the summaries' levels
2026-10-08	Residual network, calibration clause, pooled bootstrap, accelerometer series (sections 4.7, 4.8, 5.5, 5.11, 5.12)	Residual network not useful by the rule (integrated Brier -0.00008, a tenth of the required gain); series network no better than eight summaries

Date	Milestone	Outcome
2026-10-07	First delegated experiment task (spline Cox)	Four infrastructure defects found and repaired before the model could be judged: reasoning block spent the output budget (F-012), identical-probe loops under greedy decoding (F-014, F-015), server context overflow on long conversations (F-016), and the chain could not read old runs (F-013); see 5.1

Appendix B. Audit of an external literature review

An external review of the literature and of our earlier results was supplied during this work. Its statements are treated as hypotheses: each is checked against the code, the data or the arithmetic before it changes anything. This appendix records the outcome. “Checked” means we read the code or recomputed the number ourselves; “not checked” means we only have the review’s word.

Statement of the review	Status	What we found
Power: a true difference of 0.001 in IBS needs roughly 2.0 to 2.7 times our present effective information, given the observed paired standard error	Checked (arithmetic)	Reproduced: a two-sided p of 0.05 to 0.09 for a difference of 0.001 gives a standard error of 0.00051 to 0.00059; 80% power at alpha 0.05 needs 0.000357. The ratio of variances is 2.0 to 2.7. It is an approximation, since the p-values come from a corrected resampled test, not from a bootstrap.
Our p-values treat the 25 folds as independent	Checked, refuted	The existing paired comparison uses the corrected resampled t-test (variance inflated for the overlap between training sets). The review’s concern is valid for an uncorrected test and does not describe this analysis. A subject and PSU bootstrap on pooled out-of-fold predictions remains the stronger check and is reported in Results.

Statement of the review	Status	What we found
Calibration code of the evaluation library (checked by reading it, 2026-10-07)	Checked	calibration::at_horizon takes the observed risk of the whole sample and of each equal-weight risk group from an Aalen-Johansen estimate, so a subject censored before the horizon is never a non-event. Its intercept and slope are a logistic regression on the subjects whose outcome by the horizon is known, weighted by the sampling weight over the censoring survival G at the event time (or at the horizon for a subject still event-free); a subject censored earlier drops out and the weights stand in for it. G is the marginal Kaplan-Meier curve of the training data supplied by the caller, so the correction is valid when censoring does not depend on the predictors; here censoring is administrative and depends on the survey cycle, which is why every horizon metric is restricted to the cycles whose survivors were all followed that long. auc::at is the cumulative/dynamic time-dependent AUC with a competing event excluded from the controls, and says so in its documentation. The cause-specific observed risk is Aalen-Johansen, never one minus Kaplan-Meier. Gap found: the library reported no summary of the calibration curve itself (integrated calibration index, E50, E90), which is what should be judged for a step-function recalibration where the slope is not interpretable; added through the loop (section 5.4).

Statement of the review	Status	What we found
Our expected calibration error may count subjects censored before the horizon as negatives	Checked, does not apply	In the implementation the observed risk of each equal-weight risk group is an Aalen-Johansen estimate, so a subject censored before the horizon stays in the risk set until censoring and is never a negative. Slope and intercept use inverse-probability-of-censoring weights and drop subjects censored before the horizon. The censoring distribution is the marginal Kaplan-Meier of the training data; censoring here is administrative (the end of the mortality linkage) and depends on the survey cycle, which is why every metric at a horizon is restricted to the cycles whose survivors were all followed that long.
A recalibration slope after isotonic or Venn-Abers calibration does not mean what it means for a smooth score	Not checked in code; accepted as a caution	Where we report calibration after isotonic-type recalibration we report the curve, the intercept and the horizon Brier score and do not rely on the slope.
Deep survival models beat linear models at the scale of hundreds of thousands to millions of patients and billions of events; no published threshold exists	Partly checked	The scale figures were checked against the primary sources on 2026-10-07: MOTOR pretrains a 143M-parameter model on up to 55M patient records and 9B clinical events and reports a 4.6% improvement in time-dependent C-statistic over the state of the art across 19 tasks and up to 95% better label efficiency; SurvivEHR is trained on over 7.6B coded events from 23M patients; Delphi-2M trained on 402,799 UK Biobank participants and found about 2 million parameters optimal for that dataset. That no sample-size threshold exists was not independently established; we treat it as an open question and measure our own learning curve.
NHANES mortality discrimination near 0.90 is close to a ceiling for one examination	Not checked	Consistent with our own measured values. Not independently verified in the literature by us.
Gated delta-net mixers have no evidence of benefit on three to twenty sparse visits	Not checked	Our own measurement found no resolvable difference to attention on synthetic data. The line of work is frozen.

Licence and access terms of the listed cohorts

Not checked

Taken as a checklist for the data owner to confirm in writing. Nothing was downloaded or requested on this basis.

Appendix C. Label definitions for the secondary estimands

Thresholds are those of the cited guidelines (ADA for glucose and HbA1c, JNC 7 and ACC/AHA for blood pressure, KDIGO for kidney markers, WHO for anaemia and bone density). The codebooks of each cycle were used to verify variable names, units and codes only. A refused or “don’t know” answer gives a missing label, never a negative; a condition a cycle did not measure is absent for that cycle.

Label	Definition	Notes from the codebooks
diabetes	DIQ010 = 1, or DIQ050 = 1, or DIQ070 (DID070 in 2005) = 1, or HbA1c $\geq$ 6.5%, or fasting glucose $\geq$ 126 mg/dL with fasting $\geq$ 8 h	Glucose exists on the fasting subsample only and needs the fasting weights
hypertension	mean systolic $\geq$ 140 or diastolic $\geq$ 90, or BPQ050A = 1	130/80 as a sensitivity analysis
kidney markers	eGFR (CKD-EPI 2021, race-free) $<$ 60, or urine albumin/creatinine $\geq$ 30 mg/g	Creatinine variable differs in 2001 (LBDSCR); one measurement, not the three months KDIGO requires
anaemia	haemoglobin $<$ 13 g/dL (men), $<$ 12 (women not pregnant)	
high cholesterol	total cholesterol $\geq$ 240 mg/dL, or BPQ100D = 1, or BPQ090D = 1	BPQ080 is skipped when cholesterol was never checked
osteoporosis	OSQ060 = 1, or femoral-neck T-score $\leq$ -2.5 at age 50 or over	Not asked in 2011 and 2015; bone density only in some cycles
depression	PHQ-9 $\geq$ 10	2005 onward
sleep problem	SLQ050 = 1	2005 onward
undiagnosed (diabetes, hypertension, kidney markers, high cholesterol)	the measured criterion above, among those whose self-report of the diagnosis is not “yes”	“Never checked” counts as not diagnosed for cholesterol

Appendix D. Subjects per label

From `lifecourse_labels` on the 56,253 subjects. Unknown is a refused or unanswered question, a condition the cycle did not measure, or (for the undiagnosed labels) a subject outside the screened population.

Label	Positive	Negative	Unknown
diabetes	8,049	45,568	2,636
diabetes, fasting subsample	3,826	20,109	32,318
hypertension	22,257	31,943	2,053
hypertension, 130/80	28,441	25,816	1,996
kidney markers	8,413	34,839	13,001
anaemia	4,741	47,139	4,373
high cholesterol	20,620	29,180	6,453
osteoporosis (age 50 and over)	2,415	34,770	19,068
depression	3,164	33,030	20,059
sleep trouble	9,886	30,494	15,873
undiagnosed diabetes	1,335	45,568	9,350
undiagnosed hypertension	3,896	31,943	20,414
undiagnosed kidney markers	6,097	31,990	18,166
undiagnosed high cholesterol	3,421	29,180	23,652

Deaths with a diabetes or hypertension mention: 971 and 1,325 of 8,355; one death has no multiple-cause data.

Appendix E. Training details

Everything a model was given and how it was fitted, from the code at the revisions of section 8. Software: scikit-learn 1.9.1, scikit-survival 0.28.0, NumPy 2.5.3, SciPy 1.18.1, PyTorch 2.14.1 (CUDA 13), lifelines 0.30.3, pandas 2.3.3; the survival and screening arithmetic is brain’s `survival` crate. Training is unweighted everywhere; every metric is survey-weighted. The seed of fold k of repeat r is $20261006 + 100r + k$ for the Python baselines.

Shared preprocessing (all Python models, fitted on the training subjects of the fold only). Numeric inputs: missing values replaced by the training median (zero for a column missing throughout); a column that is non-negative and has a training skewness above 2 is replaced by $\log(1 + x)$; each is then standardised by the training mean and standard deviation and clipped to plus or minus 6. A missingness indicator is added for each column that has a missing value in the training subjects. Categorical answers are one-hot encoded over the levels present in the training subjects, with a level for a missing answer. The inputs of a model set are the **all** set (every examination concept, the eating-time features, the years since each recalled diagnosis and the recalled weights), the **standard** set (age, systolic and diastolic pressure, total and HDL cholesterol, HbA1c, body-mass index, sex, smoking and the told-diagnosis answers for diabetes, heart attack, coronary disease, stroke and heart failure), or age and sex.

Cox family. Elastic-net Cox (scikit-survival `CoxnetSurvivalAnalysis`): l1 ratio in $\{0.01, 0.5\}$, a path of 40 penalties down to $1e-4$ of the largest, at most 100,000 iterations, tolerance $1e-7$. A seeded 20% share of the training subjects is held out; the l1 ratio and the penalty with the best partial likelihood on it are chosen, and the model is refitted on all training subjects along the path down to that penalty. The baseline hazard is Breslow’s; the cumulative incidence at years 1 to 15 is $1 - \exp(-H_0(t) \exp(lp))$. The cause-specific model fits one such model per cause, the others censoring, on a monthly grid, and joins them by the Aalen-Johansen formula. The spline Cox model replaces each continuous column by a cubic B-spline basis with five knots at quantiles of the training column (the end ones included) and constant extrapolation outside the training range; indicator and one-hot columns pass through.

Other baselines. Inverse-weighted logistic: a logistic regression for death by 5, 10 and 15 years (or each year), each subject weighted by the inverse of the Kaplan-Meier probability of remaining uncensored, fitted on the cycles followed that long; the inverse penalty is chosen from $\{0.001, 0.01, 0.1, 1\}$ on the validation share (3,000 iterations at most), and the curve is joined by a monotone interpolation. Gradient-boosted survival (`GradientBoostingSurvivalAnalysis`, Cox loss): at most 300 trees of depth 3, learning rate 0.1 (0.25 for the fast variant), subsample 0.5, at least 50 subjects per leaf, half the features per split; the number of trees is chosen on the validation share.

Deep encoder and additive model (brain’s `horizon`, `TimelineSpec`). Hazards are piecewise exponential on 16 knots (0, 0.5, 1, 2, 3, 4, 5, 6, 7, 8, 10, 12, 14, 16, 18, 21 years). At most 96 tokens per subject, batch 256, at most 6,000 steps, peak learning rate $1e-3$ with a warm-up of one fortieth of the steps and a decay to a tenth, weight decay 0.1 (fixed by brain), evaluation every 100 steps, patience 8 evaluations, the best checkpoint kept. One subject in ten (by a hash of the seed and the id) is held out for early stopping and never trained on. The default encoder has width 64, two layers, four heads, feed-forward width 128, rank 32, 16 value bins and 8 time bins, a masked-value objective at mask rate 0.3 and weight 0.2. The additive model is the same pipeline with the interactions removed (about 16 thousand parameters). The age-sex and standard arms are the additive model on those inputs. The learning-curve recipes change only the share of training subjects, taken as a hash of the seed and the id so that a smaller share lies inside a larger.

Residual networks are specified in section 4.7 (and 4.8 for the series network): partial likelihood with Breslow ties, AdamW at $1e-3$, weight decay $1e-4$, dropout 0.1, at most 300 full-batch epochs in double precision for the tabular network (single precision for the series network), the number of epochs chosen on a 20% validation share and the model refitted on all training subjects. The series network is a convolution (2 to 16 channels, width 9, stride 2), a rectifier and dropout, a second convolution (16 to 32 channels), a rectifier, global mean and maximum pooling, and a linear head with zero weights.

Secondary estimands. Flag models: one elastic-net Cox model for death with the mention and one for death without it, per flag, as above; the share model is a logistic regression with an L2 penalty (six strengths, three-fold cross-validation by log-loss, 2,000 iterations) on the training decedents with multiple-cause data. Condition models: the logistic models use the same cross-validated penalty; the tree models are scikit-learn `HistGradientBoostingClassifier` with 300 iterations, learning rate 0.05, early stopping on a 15% share and

the fold’s seed; the age-sex and age-BMI references are logistic regressions with the library’s default L2 penalty (strength 1, 1,000 iterations). Models are fitted on the training subjects whose label is known. The screening threshold is the lowest score with at least 90% specificity among the training subjects on cross-fitted predictions (three stratified folds). The Gompertz life tables are fitted per sex by weighted maximum likelihood with delayed entry on the training subjects of the fold.

The adapter of section 5.4. Base Qwen/Qwen3-8B, a low-rank adapter of rank 16 and scaling 32, 30 optimiser steps of two records each, seed 1, weight decay 0, the base held at full precision, no-think mode, the loss on the single assistant turn that carries the accepted patch; the learning rate was not set and is brain’s default for its low-rank fine-tune (3e-4).

Appendix F. Cohort sizes used, and what a proper study would need

What this study used. Counts are from the cached feature matrix of the non-locked subjects (`baselines/_features.npz`) and the partition file.

Quantity	Value
Subjects linked to mortality	56,253 (8,355 deaths)
Locked test set, scored once under an earlier pre-registration	8,437 subjects
Subjects in cross-validation	47,816 (7,103 deaths, 14.9%)
Deaths by cause group	cardiovascular 2,218; cancer 1,572; other 3,313
One test fold	about 9,560 subjects and 1,420 deaths
One training set	about 38,200 subjects
Examinations per subject	one
Follow-up	median 9.4 years, maximum 20.8; by cycle from 19.5 years (1999) to 2.0 (2017)
Subjects whose cycle reaches 10 years	23,097 (3,266 deaths in repeat 0)
Subjects whose cycle reaches 15 years	8,976 in the pooled analysis
Subjects with an accelerometer record	10,332
Input columns of the spline Cox model	51 continuous and 16 categorical inputs, 465 columns after the spline bases
Deaths per input	about 106 (7,103 over 67)

Two consequences follow. Every comparison between models rests on the 1,420 deaths of a fold and the pooled 7,103; the learning curves of section 5.3 show the Cox models flat from half the training subjects and the flexible models still improving, which is the signature of a sample that is large for a linear model and small for a network. And the follow-up is short for the recent cycles: 2.0 to 5.9 years for the three most recent cycles of the ten, so a ten-year result uses less than half of the subjects.

How precisely the data resolve a difference. The interval of the integrated Brier score difference between the deep encoder and the spline Cox model is +0.00019 to +0.00204, half a width of 0.0009, at 7,103 deaths. If the standard error falls with the square root of the number of deaths and the correlation between the two models’ errors stays as it is here (an assumption, not a measurement), the number of deaths needed so that the 95% interval is narrow enough to detect a true difference of size D with 80% power (half width 0.7 D) is:

True difference in IBS to detect	Deaths needed	Subjects at this cohort’s 14.9% death share
0.0010	about 12,000	about 83,000
0.0005	about 49,000	about 330,000
0.0002	about 300,000	about 2,000,000

These are scaling arguments from one comparison in one cohort, to plan the order of magnitude; a protocol would compute power from simulation on its own data. They are consistent with the scale at which others report gains (section 2: hundreds of thousands to millions of patients).

What a longitudinal comparison needs. The question of whether many examinations per person beat the single-examination models cannot be answered with this cohort, which has one examination per subject. A proper study needs:

1. *Repeated measurements of the same markers*, at least three examinations per person over at least ten years, with the same laboratory methods, so that trends can be separated from assay change.
2. *Outcomes beyond death*, linked from hospital, cancer and prescription registers, with at least 2,000 incident events for each outcome to be modelled (diabetes, myocardial infarction, stroke, cancer, kidney disease) and the date of onset, not a certificate mention.
3. *Size*, at least 12,000 deaths (about 80,000 persons of this age mix) to resolve a gain of 0.001 in the integrated Brier score, and several times that if the gain is expected to be smaller than 0.001.
4. *Comparators fixed in advance*: age and sex alone, the conventional risk score for the outcome (for example SCORE2 or QRISK3 for cardiovascular disease, a biological-age clock for death), and the single-examination spline Cox model of this paper, each given the same most recent examination, so that the question is what the history adds.
5. *Locked temporal and external test sets*: the most recent examinations held out in time, and a second cohort from another country or health system, scored once, with calibration reported before discrimination.
6. *The statistical design of this paper*: rules fixed before results, folds that respect the person (all examinations of a person in the same fold), cluster bootstrap over the sampling units, and a report of every model tried.

Candidate data sets, counts as published by their owners and not verified beyond them: the UK Biobank (about 500,000 participants, with 20,323 repeat assessments in 2012 to 2013 and 21,469 in the first repeat imaging visit, which is small for this purpose); the Whitehall II cohort (10,308 civil servants, examinations at alternate phases over more than 30 years, which has repeated markers but few deaths per marker); and electronic health records of a health system, where examinations are irregular and the size is in the millions. No single one of these both has repeated markers and the 12,000 deaths needed; registry-linked primary-care records are the most plausible route to the size, and cohort studies to the measurement quality. All require an application or an agreement that only the owner of this work can make.

What the variable questions need. The importance analysis of section 5.13 describes what the model uses. To say what would happen if a person changed something, the data must record the change and the reason for it. That is a randomised trial, or a target trial emulation on longitudinal data with methods for time-varying confounding (inverse-probability weighting or the g-formula), a pre-specified intervention (for example starting a statin, or reaching a given blood pressure), a confounder set measured before the decision, and a sensitivity analysis for unmeasured confounding. Even then, the result is an estimate for a population of the cohort's kind, to be read by a clinician, not advice for an individual.